

Outside Options and Collective Bargaining Spillovers

Luis de Azevedo Gomes

Guilherme Neri

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Abstract

This paper studies how stronger unions generate spillovers through labor market competition. We exploit a Brazilian reform that strengthened union bargaining at a subset of firms and identify competing employers from workers' job mobility networks. Directly affected firms raise wages and union-bargained amenities. Competing employers raise wages by one-quarter as much, while amenities are unchanged. We find no evidence that unions actively mediate these spillovers. Employers instead respond to competition, raising wages most where they compete for costlier-to-replace workers, and leaving workforce size and composition largely unaffected. Within firms, pay does not respond to outside-option heterogeneity, suggesting internal wage-setting constraints.

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Kellogg School of Management, Northwestern University. Emails: luis.gomes@kellogg.northwestern.edu, guilherme.neri@kellogg.northwestern.edu. We thank Abi Adams and Ben Friedrich for invaluable guidance and extensive feedback. This paper benefited from comments by Luis Alvarez, José Antunes Neto, Lori Beaman, Netanel Ben-Porath, Nicola Bianchi, Daniela Borja Arriaga, Michael Powell, Myera Rashid, Chris Udry, and from participants at the Kellogg Strategy Brown Bag and Northwestern's HELP Workshop. We thank Lorenzo Lagos for making the CBA registry data available, and for answering questions about the data and setting.

1 Introduction

Firms do not set pay in isolation. When labor markets are imperfectly competitive, workers' compensation depends not only on productivity but also on their outside options.^{1,2} Labor market policies and institutions can therefore have indirect effects. By changing compensation at some workplaces, they shift the outside options of workers elsewhere, and with them the pay competing employers must set to attract and retain workers. The effect of a policy thus reflects not only how it bears on each firm directly, but also how it reshapes competition in the labor market.

Collective bargaining governs the pay of a large share of workers around the world (Jäger, Naidu and Schoefer, 2025). While imperfect competition allows firms to capture labor market rents, unions can shift part of those rents to covered workers, especially where they hold greater bargaining power (Dodini, Stansbury and Willén, 2024; Lagos, 2026).³ Whether stronger unions also affect workers in the broader labor market remains far less clear.

We study how union bargaining power at some firms shapes compensation at competing employers. Our setting is Brazil, where the formal workforce is universally represented by unions, assigned administratively rather than through worker organizing. We exploit an unexpected legal reform to the rules governing collective bargaining agreements (CBAs), which are negotiated between unions and employers on a recurring basis. Before the reform, CBAs expired at the end of the contract term, allowing employers to revert to statutory labor law minima. This gave employers a credible holdout threat, since the possibility of delaying a new agreement past expiration pressured unions into concessions. In September 2012, Brazil's Superior Labor Court eliminated this leverage, mandating that existing CBA terms remain in effect until a new agreement is signed.

Our identifying variation in union strength comes from which establishments happened to hold an active CBA when the reform was approved.⁴ Establishments with an active CBA at the

¹Different labor-market frameworks allow for prospective employers to affect wages and employment (e.g. Burdett and Mortensen, 1998; Cahuc, Postel-Vinay and Robin, 2006; Caldwell and Danieli, 2024).

²A growing literature provides evidence of labor market power across many contexts (e.g. Azar, Marinescu and Steinbaum, 2022; Lamadon, Mogstad and Setzler, 2022; Sharma, 2023).

³Azkarate-Askasua and Zerecero (2025) formalizes this logic, with the wedge between wages and the marginal revenue product of labor reflecting both the firm's wage markdown and the union's bargaining share, so that greater union bargaining power translates into a larger share of rents accruing to workers.

⁴This setting was first exploited by Lagos (2026).

time, the directly treated, entered subsequent negotiations with their existing agreement locked in as the new disagreement point. Establishments without an active agreement at the time, the untreated, had no terms for the reform to preserve, so the employer’s holdout threat of reversion to statutory minima remained intact. Because each negotiated agreement becomes the baseline for subsequent rounds, this initial asymmetry can generate persistent differences in bargaining outcomes.

We combine matched employer-employee records with a registry of collective bargaining agreements. The employer-employee records (RAIS) cover the near-universe of formal employment relationships, reporting wages and employee characteristics, along with formal work histories. The CBA registry (Lagos, 2026) records every agreement negotiated, identifying its content and the unions and establishments involved in each. It allows us to assign treatment status, to measure collectively bargained amenities from the clauses of each agreement, and to map which establishments share a union.

To consider spillovers, we proxy for workers’ outside options with realized job transitions between establishments, a revealed-preference measure we call connectivity.⁵ Rather than defining the relevant labor market *ex ante*, we let workers’ transitions characterize it. While industry and location predict connectivity, they explain a minority of its variation. The remainder is not merely sampling noise, since connectivity in earlier years predicts connectivity in later years, even conditional on observables. Our measure thus captures durable competitive relationships between employers that standard classifications miss. For each untreated establishment, we measure indirect exposure to the reform with total pre-reform connectivity to directly treated establishments.

The reform raised both wages and collectively bargained amenities at directly treated establishments, where unions gained bargaining power. Our difference-in-differences design compares them with establishments that also negotiated frequently but held no active CBA when the reform was approved. To account for potential spillovers, we require the control group of untreated establishments to have zero connectivity to the directly treated. Wages increased by 2.8% and subsequent CBAs added 1.6 clauses, 5% of the pre-period mean, with no detectable employment effects. Excluding connected establishments from the comparison group raises the estimated direct effect on wages by about 40%, in the direction implied by positive wage spillovers.

⁵Our approach follows recent literature in leveraging cross-firm worker transitions to measure labor markets and estimate spillover effects (e.g. Arnold, 2022; Nimczik, 2023; Bassier, 2024; Demir, 2025; Derenoncourt and Weil, 2025).

Competing employers raised wages in response, though amenities did not change. Employees of highly connected untreated establishments gained 0.65% in wages, about a quarter of the direct effect. We detect no spillovers on employment. These estimates are robust to flexible controls for establishment characteristics, workforce composition, and local industry exposure to the reform, with the ratio of spillover to direct wage effects ranging from 0.20 to 0.26 across specifications. To separate exposure to the directly treated from general connectedness, we follow Borusyak and Hull (2023) and control for connectivity to counterfactual treated sets, formed by randomly reassigning treatment status among comparable establishments. Counterfactual exposure has no effect on wages, and the spillover estimate holds once we control for it. The wage response therefore rests on ties to the directly treated specifically.

While unions raised pay where they gained bargaining power, they did not mediate spillover effects. The canonical confounder to the outside-options interpretation is the unionization threat effect: employers raising pay to deter union formation (Freeman and Medoff, 1981; Lewis, 1986; Fortin, Lemieux and Lloyd, 2021). Universal, administratively assigned representation rules it out by construction, but union channels of a different kind remain. Since most untreated establishments share a union with directly treated employers, that union might expand its bargaining capacity through higher revenues, or reallocate effort across the establishments it represents. Any such channel would operate through what unions produce, their negotiated agreements, which the CBA registry records. We find no evidence of bargained clauses responding to connectivity in counts, composition, or wage-equivalent value, and the wage spillover is robust to union-specific controls. These findings suggest a broader principle. A labor market institution can generate spillovers through its direct effects on compensation alone, propagating through competition for workers rather than through the institution itself.

We then ask how else employers respond to competition, and find that wage increases left workforce churn and composition largely unchanged. While a null on employment could still mask adjustment on other margins, we detect no substantial effects on separations, hiring, and contracted hours. Turning to the firm-level wage gains, whether they stem from a shift in workforce composition or from a higher price of labor is not obvious a priori. To separate the two, we estimate each establishment's average deviation of wages from what observationally equivalent workers earn nationwide, predicted by a wage regression on worker characteristics estimated across the formal sector. Most of the spillover wage effect reflects a higher price of labor, with

the remainder consistent with modest workforce upgrading.

Having documented spillovers across firms, we examine whether the same outside-option channel operates within them. Because our connectivity measure is built from worker-level mobility flows, we can compute it separately for subgroups of workers within a single employer, so that two groups at the same firm may face different exposure to directly treated establishments. Partitioning workers by education, gender, and tenure, we test whether the more-exposed workers experience larger wage gains than their coworkers. They do not. Firms thus show little sign of tailoring pay to group-specific outside options, pointing to constraints on internal pay differentiation. Across firms, however, the wage response is larger where connectivity runs through longer-tenured workers, consistent with firms adjusting pay most where workers are costlier to replace.

Literature. Our results relate to frameworks of labor markets with imperfect competition, where wages are shaped by workers’ outside options (Kline, 2025).⁶ Empirical work documents this link, exploiting variation in the value of non-employment (Nekoei and Weber, 2017), beliefs about market wages (Jäger et al., 2024), labor market concentration (Schubert, Stansbury and Taska, 2024; Arnold, 2025), and local labor demand (Caldwell and Harmon, 2019). Closer to this paper, previous research estimates wage spillovers from policies at specific firms or sectors (Staiger, Spetz and Phibbs, 2010; Willén, 2021; Derenoncourt and Weil, 2025).

We provide evidence of compensation spillovers considering not only wages but also a measure of amenities, a margin largely absent from previous analyses.⁷ We combine variation in union bargaining power, which directly affects a broad set of compensation margins, with CBA data containing information on bargained clauses. While employers directly affected by the reform raise both wages and amenities, spillovers appear in wages only, with no detectable change to the number, type, or value of amenities at competing firms.

We also contribute to a literature on heterogeneity in workers’ outside options within the firm. Previous work documents outside options varying substantially across workers, accounting for

⁶Equilibrium search models with on-the-job search provide one microfoundation for these dynamics: the possibility of poaching disciplines firms’ wage setting and generates wage dispersion, both under wage posting (Burdett and Mortensen, 1998) and under wage bargaining (Postel-Vinay and Robin, 2002; Cahuc, Postel-Vinay and Robin, 2006).

⁷Relatedly, Park, Stainier and Stansbury (2025) provides evidence of better outside options reducing workplace injuries.

a meaningful share of wage inequality (Sharma, 2023; Caldwell and Danieli, 2024). Our results speak to how finely firms tailor wages to workers’ outside options, since indirect exposure to the studied reform varies across worker groups within the same establishment. We find no differential wage response. Dispersion in outside options within the firm therefore need not translate into dispersion in pay, consistent with studies on internal wage-setting constraints (Manning, 1994; Dube, Giuliano and Leonard, 2019; Amior and San, 2026). This also informs the measurement of labor market spillovers, as exposure defined below the firm level introduces variation employers may not flexibly price, attenuating the estimated response.

An extensive literature studies the effects of unions and collective bargaining, mostly considering the effects of unionization (e.g. DiNardo and Lee, 2004; Lee and Mas, 2012; Frandsen, 2021). Union bargaining power conditional on union coverage may also shape the compensation structure in the labor market, including in contexts where representation is universal.⁸ Lagos (2026) studies this margin in the same setting we examine, showing that workers covered by stronger unions experience increases to both wages and union-bargained amenities. We document that compensation gains from stronger unions extend to the broader workforce, but only through wages, with no change to amenities.

Prior research examines the spillover effects of unions, focusing primarily on how unionization affects non-covered workers through threat effects — non-union employers raising wages to deter unionization (Freeman and Medoff, 1981; Lewis, 1986; Fortin, Lemieux and Lloyd, 2021).⁹ Current work studies spillovers from wage floors set by sectoral bargaining, tracing effects through workers’ job mobility (Bassier, 2024; Demir, 2025).¹⁰

We add to this literature in two ways. First, we study spillovers from union bargaining power rather than from unionization or wage floors. Second, our setting and data allow us to test directly whether an institutional channel drives spillovers. Where unions arise through worker organiz-

⁸The literature on unionization mostly studies “Wagner Act” systems (the U.S. and Canada), in which coverage is determined by establishment-level certification and so nearly coincides with union membership. Such systems are not common: elsewhere, sectoral bargaining and mandatory extensions decouple the two, and coverage can be widespread even where union density is low, as in Brazil, France, and Spain (Jäger, Naidu and Schoefer, 2025).

⁹While Fortin, Lemieux and Lloyd (2021) frames its evidence on spillover effects as resulting from unionization threats, it acknowledges the possibility of an outside-option channel consistent with imperfect competition in the labor market.

¹⁰The two settings differ in the adjusting margin. Bassier (2024) studies increases in wage floors set by South African bargaining councils and finds that non-covered local industries exchanging more workers with covered firms raise wages. Demir (2025) studies the announcement of Germany’s first sectoral minimum wage and finds wage gains for connected workers driven by moves to better-paying employers rather than by wage responses at more exposed establishments.

ing, spillovers to non-union firms blend competition for workers with threat effects, which are difficult to separate empirically.¹¹ Universal, administratively assigned representation removes this confound by construction. It admits union-mediated channels of its own, for which our data on bargained content and union coverage show no evidence. Competition between employers can thus be sufficient to transmit the effects of unions.

A separate literature asks whether competition affects union activity itself, including unionization, elections, and strikes (Ahlquist and Downey, 2023; Pezold, Jäger and Nüß, 2023; Charles, Johnson and Tadjfar, 2026). We instead study whether competition in the labor market affects the content and value of union-bargained contracts. Our evidence suggests it does not.

Outline. The remainder of this paper proceeds as follows. Section 2 reviews the collective bargaining setting in Brazil. Section 3 describes the data and sample restrictions. Section 4 presents our measure of outside options. Section 5 outlines the empirical strategy and results. Section 6 examines additional margins of firm adjustment and outside options within the firm. Section 7 concludes.

2 Institutional Context

In this section, we describe the union landscape in Brazil and its collective bargaining system. We then detail the reform used to construct the union bargaining power variation.

2.1 Unions in Brazil

Collective representation in Brazil is universal: every formal worker is represented by a union, assigned through government-determined coverage rather than through establishment-level organizing.¹² A single union holds exclusive rights to represent all workers within a given professional category and geographic jurisdiction. Professional categories do not map cleanly into standard industry or occupation classifications, while geographic jurisdictions range from individual

¹¹Green et al. (2023) addresses the same question within a structural model, allowing unions to affect non-union wages through both the threat channel and a bargaining channel in which non-union workers value the prospect of moving to a union job. Estimating the model on U.S. data, it attributes nearly all of the spillover from de-unionization to the bargaining channel.

¹²For a comparative discussion of union landscapes, see Jäger, Naidu and Schoefer (2025).

municipalities to broader groupings, including state-wide coverage. As a result, the category-geography cells that define union representation do not correspond to standard industry-location pairs.¹³

Assignment operates through an administrative process at the Labor Ministry, which links each establishment to its union based on primary economic activity and location. Neither firms nor unions exercise choice over this assignment, which eliminates scope for strategic sorting: firms cannot select into favorable unions, and unions cannot selectively represent particular employers.

The resulting bargaining landscape is highly fragmented, with more than six thousand unions in operation. Universal representation, however, should not be conflated with membership: workers are represented regardless of whether they join their union, and union density is relatively low.¹⁴

2.2 Collective Bargaining Agreements

Unions bargain through collective bargaining agreements (CBAs), which are legally binding contracts governing wages, benefits, working hours, and other terms of employment. CBA provisions supersede individual contracts and may only add to statutory protections established in federal labor law. When a CBA applies, workers cannot opt out of its terms, and employers must extend those terms to all workers covered by the bargaining unit.

Coverage, however, is narrower than representation. Although every formal worker is assigned to a union, at any given point in time many workplaces are not covered by an active CBA. CBAs are temporary contracts, with a legal maximum of two years and a typical term of twelve months, so maintaining coverage requires parties to renegotiate before each agreement expires. Unions exercise discretion over whether and when to initiate these negotiations. As a result, some negotiate infrequently, others forgo negotiations with certain employers, and many agreements lapse without renewal. The recurring renegotiation cycle also yields repeated observations of bargaining outcomes within the same union-employer relationships.

CBAs are negotiated at two levels. Sectoral CBAs are agreements between a union and an

¹³The average union represents workers across 13 industries and 7 microregions (Derenoncourt et al., 2025).

¹⁴Union membership in Brazil is approximately 15% and is not required for CBA coverage.

employer association representing all firms in a category-geography cell. These agreements establish floors and standards that apply across multiple establishments. Firm-level CBAs are agreements between a union and individual employers. These build upon sectoral agreements to provide establishment-specific provisions tailored to particular workplaces (Horn, 2009).

Since 2009, the Brazilian Ministry of Labor has required all CBAs to be registered in *Sistema Mediador*, an online platform, to gain legal validity. This administrative mandate generates a comprehensive registry of CBA texts which includes information on CBA coverage, negotiation dates, and contract provisions.

2.3 The 2012 Ultractivity Reform

Before September 2012, CBA provisions expired at the end of the contract term. If parties had not negotiated a replacement agreement by the expiration date, employers were not bound by the terms of the expired CBA, being able to legally revert to statutory labor law minima. This gave employers leverage during negotiations: by allowing an agreement to lapse, they could pressure unions into concessions. Since bargaining often extended beyond formal expiration dates, this leverage was a recurring feature of the negotiation process.

On September 25, 2012, the Superior Labor Court (*Tribunal Superior do Trabalho*, TST) revised *Súmula 277*, introducing *ultractivity*: the legal principle that expired CBA provisions remain enforceable until parties sign a replacement agreement. The revision applied to all clauses in existing CBAs that had not yet expired. While this revision represented a reversal of the court's prior interpretation and had limited legal precedent, it was issued during a regular session with minimal advance notice.

Ultractivity shifts the disagreement point in collective bargaining. Under the previous regime, a negotiation impasse meant that CBA provisions could lapse, leaving statutory minima as the fallback. Under ultractivity, the terms of the expired agreement continue to bind the employer, allowing the union to hold out on existing conditions. Because CBAs may only expand upon statutory protections, the union's disagreement point under ultractivity is weakly higher than under the previous regime.

The timing of when unions first benefit from this shift creates the identifying variation we exploit. Establishments with an active CBA in September 2012, the directly treated, enter their

next negotiation with ultractivity protecting provisions negotiated before the reform. Establishments without active coverage at the time of the reform, the untreated, must first negotiate a new agreement starting from statutory minima, since there are no existing provisions for ultractivity to preserve. Because each negotiated agreement becomes the disagreement point for the subsequent round, this initial asymmetry can generate persistent differences in bargaining outcomes over time (Lagos, 2026).¹⁵

3 Data

Relação Anual de Informações Sociais (RAIS). RAIS is a matched employer-employee administrative dataset covering the near-universe of formal employment relationships in Brazil. Maintained by the Ministry of Labor, it is compiled from mandatory annual filings by all registered establishments, providing high coverage and reliability.

Each establishment is identified by a unique tax identifier (CNPJ), with records of its industry and municipality. Each worker carries a unique identifier that allows tracking across employers and over time. The data include earnings, contracted hours, and demographic characteristics, including age, gender, race, and education. We define the monthly wage as earnings in December and the hourly wage as those earnings divided by contracted monthly hours.¹⁶ Hourly wages are our main outcome, since they hold measured pay fixed against adjustments along the hours margin. We report monthly wages alongside them in the main results, where the two deliver similar estimates.

CBA Registry (Lagos, 2026). This dataset consists of web-scraped information on the universe of collective bargaining agreements between unions and firms extracted from *Sistema Mediador*, the registry of all CBAs negotiated since 2009 in Brazil.¹⁷ It includes variables identifying the parties signing the CBAs and relevant dates (filing, start, and end). Our main amenity outcome is the total number of clauses in a given agreement. Each clause is an individual provision such as a

¹⁵ Although ultractivity applies to all agreements signed after the reform, untreated establishments need not catch up: as Lagos (2026) argues, with ultractivity now anticipated, employers concede less in these establishments' first post-reform negotiations, since any granted provision becomes downward-rigid. This logic follows the insight that anticipated downward rigidity compresses wage increases (Elsby, 2009).

¹⁶ RAIS records contracted hours on a weekly basis, following the convention in Brazil. We convert to monthly hours by multiplying reported weekly hours by 4.348, the average number of weeks in a month.

¹⁷ The CBA registry data can be accessed at <https://sites.google.com/view/lorenzolagos/data>.

maternity leave entitlement, an overtime pay rule, or a job stability guarantee, and is classified by the negotiating parties into predefined types when the CBA is filed. *Sistema Mediador* organizes these types into three broad groups—wage-related, employment-related, and other clauses—and 24 finer groups, which we use to examine the composition and value of agreements. To match CBAs with the RAIS data, we web-scraped additional information on the geographic coverage of each agreement from *Sistema Mediador*.¹⁸

Sample Restrictions. We combine these two datasets to construct an establishment-year panel spanning 2009 to 2016.¹⁹ Establishments are matched to CBAs using employer identifiers and the CBA’s geographic coverage.

We impose two sample restrictions. First, we require establishments to have negotiated at least two CBAs before 2012, including one in 2009, and at least one after. This restricts the comparison to untreated establishments that actively engage in collective bargaining but happened to be between agreements when the reform took effect, rather than firms that rarely negotiate. It also allows us to construct pre-treatment baselines for CBA characteristics, which we use to test for differential pre-trends in collectively bargained amenities. Second, we restrict to a balanced panel of establishments present in all years, so that estimates reflect changes within a fixed set of continuing establishments rather than shifts in sample composition. We also focus exclusively on firm-level CBAs. Under sectoral CBAs, exposure to the reform is common to all firms in a covered category-geography cell. Firm-level agreements generate treatment variation at the employer level, allowing us to study spillovers both within and across covered cells.^{20,21}

Direct treatment assignment follows from CBA timing. An establishment is directly treated if it had a firm-level CBA filed before September 25, 2012, with a contractual end date after that date. Since both filing and expiration dates were determined before the reform was announced, treatment status reflects pre-existing negotiation timing rather than strategic responses to the

¹⁸Geographic coverage information is necessary because a firm-level CBA may not cover all establishments of a firm. Since unions have geographic jurisdictions, a firm with establishments in multiple municipalities may be covered by different firm-level CBAs. We use the geographic coverage data to determine which establishments fall within each agreement’s scope.

¹⁹The exclusion of years after 2016 is motivated by a labor market reform passed in 2017 that reversed the 2012 ultractivity reform.

²⁰These sample restrictions closely follow those described in Lagos (2026), which studies the direct effects of the same reform.

²¹Throughout, the unit of observation is the establishment, and we use “firm” interchangeably where the distinction does not bear on the analysis.

policy change. For these establishments, ultractivity immediately locked in existing provisions as the new disagreement point. An establishment is untreated if it did not have an active firm-level CBA on the reform date. When untreated establishments next entered negotiations, no existing provisions were in force for ultractivity to preserve, leaving the employer’s initial holdout threat intact.

4 A Measure of Outside Options

We require a measure of each untreated firm’s indirect exposure to the reform, both to estimate spillover effects and to address how spillovers may bias direct effects. We focus on indirect exposure operating through workers’ outside options. When compensation changes at directly treated establishments, an untreated firm may respond to the extent that its workers view treated establishments as potential workplaces.

We measure outside options using worker movement between employers, following a similar approach to that of recent work (e.g. Arnold, 2022; Bassier, 2024; Nimczik, 2023; Demir, 2025; Derenoncourt and Weil, 2025). When transitions between two employers are frequent, they likely draw from overlapping labor pools and represent credible alternatives for each other’s employees. This revealed-preference approach captures dimensions of labor market competition that common administrative data cannot directly observe, such as skill requirements, firm-specific amenities, and referral networks.²²

Alternative approaches define labor market competition through observable classifications, such as industry codes and location.²³ These measures require the researcher to fix the boundary of each firm’s market in advance, committing to which other firms count as relevant alternatives. This is especially demanding in our setting. Treatment is dispersed across many industry-location cells, and within each affected cell may reach only a subset of firms, so cells are not uniformly treated. Measuring an untreated firm’s exposure then requires two choices, neither with a prin-

²²In other contexts, mobility patterns reflect skill requirements not captured by standard occupation or industry codes (Poletaev and Robinson, 2008; Gathmann and Schönberg, 2010; Schubert, Stansbury and Taska, 2024). Non-wage job attributes may vary substantially across firms, including within industries and locations (Sorkin, 2018; Sockin, 2022). Referral networks transmit information and facilitate matches through channels absent from administrative records (Dustmann et al., 2016).

²³Existing work uses such alternative measures to define labor markets (e.g. Prager and Schmitt, 2021; Benmelech, Bergman and Kim, 2022; Berger, Herkenhoff and Mongey, 2022; Rinz, 2022).

ciplend answer. The first is which directly treated firms to count as its alternatives. The second is how to weight those firms into a single exposure measure, since cells differ in the share of firms and of workers treated by the reform. Worker flows sidestep both choices, letting realized transitions signal competitive relationships at the establishment level without imposing boundaries *ex ante*.²⁴

Specifically, for workers in firm i , we assign firm j to be a more credible outside option if worker movements between the two firms are large as a proportion of firm i 's total employment. This measure captures two distinct margins. First, it reflects how frequently transitions between i and j occur relative to transitions between i and other employers. Second, it accounts for the overall rate of worker movement at firm i . Both margins matter when estimating average effects at the firm level: even if a large share of firm i 's mobile workers transition between i and j , firm j constitutes a weaker outside option for the typical worker in firm i if movement is rare.

We construct this measure using worker job transitions in the pre-reform period, averaged across four pairs of years (2007–2008, 2008–2009, 2009–2010, and 2010–2011). Aggregating over multiple years reduces noise from year-to-year variation in realized worker movements. Restricting to the pre-treatment period avoids potential endogeneity: if the reform altered worker mobility patterns, a measure including post-treatment flows would be confounded by treatment effects. Formally, we define the (pairwise) connectivity between firms i and j as:

$$(1) \quad C_{ij} = \frac{1}{4} \sum_{t=2007}^{2010} \frac{\text{Flows}_{ij,t,t+1}}{\text{Avg. employment}_{i,t,t+1}}$$

where $\text{Flows}_{ij,t,t+1}$ denotes the number of workers moving between firms i and j in either direction from year t to $t + 1$, and $\text{Avg. employment}_{i,t,t+1}$ is the average employment at firm i across the two years.

Pairwise connectivity C_{ij} is higher when firms i and j are geographically close, offer similar compensation (wages and collectively bargained amenities), employ demographically similar

²⁴Recent work using job mobility networks to estimate spillovers aggregates exposure to a level coarser than the individual firm. Bassier (2024), for example, measures a firm's exposure from worker flows between its industry-region cell and the covered bargaining council, and Arnold (2022) from transition rates between occupation-microregion cells. Because treatment in our setting varies within these cells, we measure connectivity at the firm level, so that exposure reflects each firm's own ties to the directly treated rather than those of its cell.

workforces, or share the same industry or location (Figure 1, Panel A). Common union coverage also predicts worker movement, even after conditioning on other factors. This pattern may reflect union jurisdictions tracking firms in related activities beyond what standard industry codes capture, or unions facilitating workers’ awareness of job opportunities at other covered employers.

Being in the same “local industry”, as measured by sharing an industry–microregion cell, markedly increases predicted connectivity. Yet such classifications account for only a minority of the variation. We regress C_{ij} on fixed effects for each pairing of the two establishments’ industry–microregion cells ($\text{industry}_i \times \text{microregion}_i \times \text{industry}_j \times \text{microregion}_j$), a specification that nests any labor-market definition based on these classifications. This regression yields an R^2 of 0.22, leaving most of the variation unexplained. This gap is not purely an artifact of sparse, low-movement pairs of employers: even among an establishment’s strongest connections, those in the top decile of establishments with which it exchanges workers, only half share an industry–microregion cell.

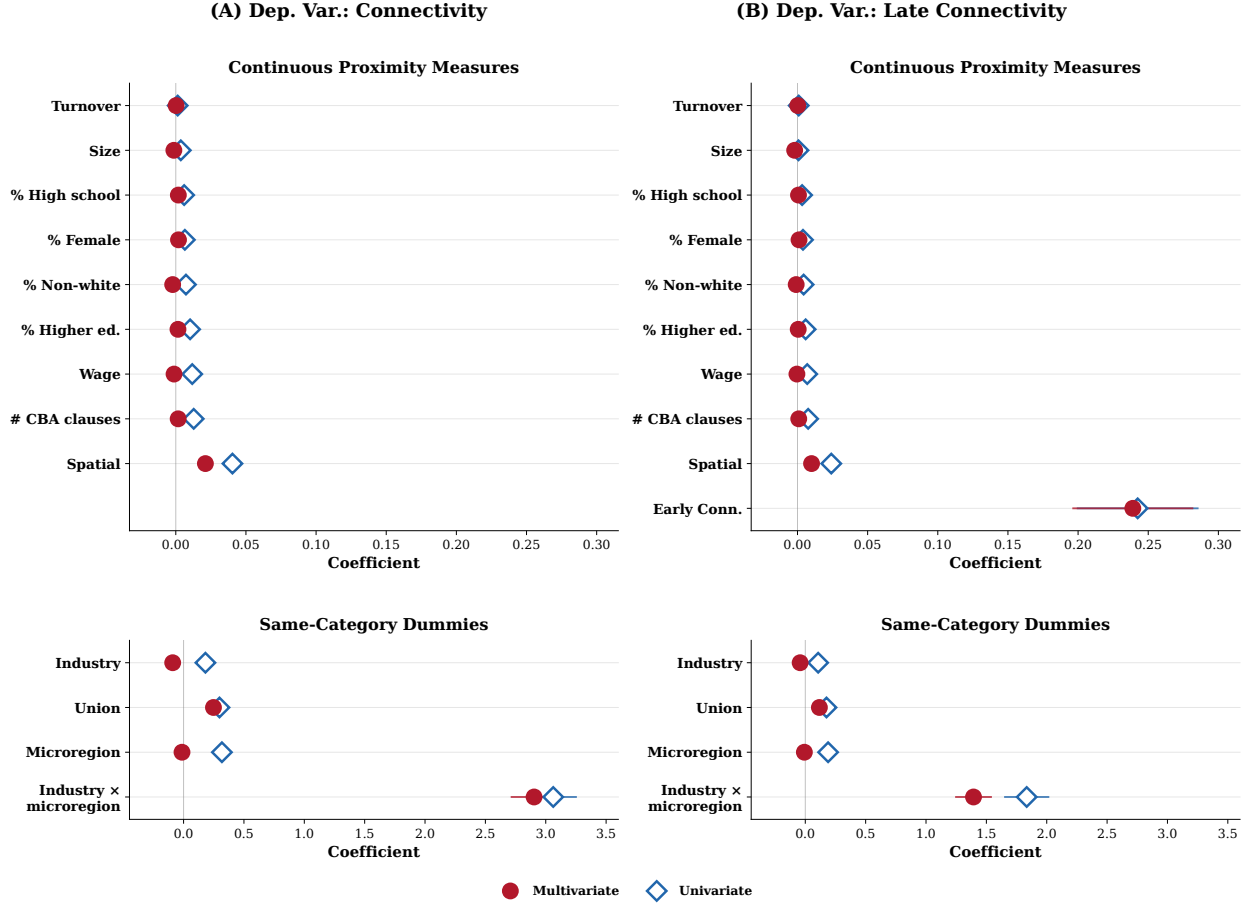
Residual variation in our measure could in principle reflect either persistent features of labor market competition or idiosyncratic noise in realized transitions. To distinguish between these possibilities, we split the estimation window into an early and a late period, and test whether early connectivity predicts late connectivity, conditioning not only on shared industry and location but on proximity in wages, firm size, worker demographics, and geographic distance. Higher early connectivity predicts higher late connectivity, even after conditioning on these characteristics (Figure 1, Panel B). Worker movements thus retain a stable, predictive component that observable measures of firm similarity do not capture, whether those measures rest on industry and location or on the broader set of characteristics a researcher might use to define a labor market.

To estimate spillover effects from the reform, we aggregate pairwise connectivity across all directly treated firms j . That is, for each untreated firm i , we define:

$$(2) \quad C_i = \frac{1}{4} \sum_{t=2007}^{2010} \frac{\text{Flows with treated}_{i,t,t+1}}{\text{Avg. employment}_{i,t,t+1}}$$

This measure proxies for the degree to which firm i ’s workers perceive directly treated establishments as outside options. Higher values indicate that firm i competes more intensively for workers with firms where unions gained bargaining power.

FIGURE 1: PREDICTORS OF PAIRWISE CONNECTIVITY BETWEEN ESTABLISHMENTS



Notes: This figure reports regressions of pairwise connectivity, defined in equation (1), on establishment-pair characteristics. Results concern the analysis sample, further described in Section 5. The unit of observation is an ordered establishment pair. In Panel (A), the dependent variable is connectivity measured over the full pre-treatment window (2007–2011). In Panel (B), the dependent variable is connectivity measured from late transitions (2009–10 and 2010–11), and early connectivity, measured from transitions 2007–08 and 2008–09, is added as a regressor. The top panels display continuous proximity measures, defined as the negative absolute difference between establishment characteristics (standardized), so that higher values indicate greater similarity. Spatial proximity is the negative of the distance between ZIP code centroids. The bottom panels display indicators equal to one if both establishments share the indicated category. Hollow blue markers show univariate regressions. Filled red markers show a single multivariate regression including all predictors, from both the continuous proximity measures and the same-category dummies. All regressions include establishment- i fixed effects. Establishment characteristics are pre-reform averages. Horizontal lines represent 95% confidence intervals, which are included for all estimates.

5 Empirical Approach and Results

In this section, we describe the empirical strategy, explaining how the analysis sample and regression specification allow for the causal estimation of the direct and spillover effects of increased union bargaining power, and discuss the main results.

5.1 Direct Effects

To estimate direct effects, we first acknowledge the possibility of bias from a violation of the stable unit treatment value assumption (SUTVA). If the reform results in spillovers, untreated firms connected to directly treated firms through competition in the labor market would themselves experience changes in outcomes. If, for example, spillovers operate in the same direction as direct effects, the latter would be attenuated.

To mitigate this concern when estimating direct effects, we restrict the control group to establishments with zero labor market connectivity to directly treated firms, that is, those with $C_i = 0$. The comparison group is thus composed of untreated firms that exchanged no workers with the directly treated over the four year-pairs preceding the reform (2007–2011). For comparison, we also report direct-effect estimates using all untreated firms as the control group. The contrast between the two shows how the estimated direct effect changes once connected firms enter the comparison group, a difference we test formally with inference that accounts for the two estimates not being independent.

Sample. Before accounting for connectivity, directly treated and untreated establishments differ in levels along several dimensions. In particular, directly treated firms pay higher wages, employ longer-tenured workers, are distributed differently across sectors, and negotiate CBAs in different months (Table [A1](#), Figure [B1](#)).

The zero-connectivity untreated firms, used as the comparison group in our main specification, are substantially smaller. This is partly a mechanical consequence of the zero-connectivity restriction: smaller firms are more likely to see no worker movement with any specific set of establishments, including the directly treated. Relatedly, per-worker flows at zero-connectivity firms are lower and average tenure is higher than at the average untreated firm. While our

difference-in-differences design absorbs level differences, these characteristics could predict differential outcome trajectories, motivating the time-varying controls introduced next.

Specification. We use a Difference-in-Differences (DiD) approach to estimate the causal direct effects of the reform on wages, employment, and collectively bargained amenities. Explicitly, we use the following specification:

$$(3) \quad Y_{it} = \alpha_t + \eta_i + \sum_{\substack{k=2009 \\ k \neq 2011}}^{2016} \beta_k (D_i \times \mathbb{1}_{t=k}) + \sum_{\substack{k=2009 \\ k \neq 2011}}^{2016} \Gamma_k (X_i \times \mathbb{1}_{t=k}) + \varepsilon_{it}$$

where Y_{it} is the outcome of interest in establishment i at year t . α_t and η_i correspond to year and establishment fixed effects, respectively.

The dummy variable D_i takes value $D_i = 1$ for establishments with a CBA filed before September 25, 2012, whose contractual end date fell after that date, and $D_i = 0$ otherwise. The interaction $X_i \times (\mathbb{1}_{t=k})$ denotes time-varying controls: fixed effects interacted with year indicators for industry, microregion, and negotiation month, together with year-interacted quartile bins of the pre-treatment outcome, firm size, and per-worker flows with all other firms. Interacting each of these dimensions with year indicators lets outcomes trend differently along them, rather than constraining establishments that differ in category or baseline level to a common trajectory. The inclusion of size and per-worker flows is further motivated by our spillover estimation approach (Section 5.2), which uses the same controls for consistency.

The coefficients of interest are β_k , for $k \in \{2009, \dots, 2016\} \setminus \{2011\}$. They capture the differential evolution of outcomes in treated versus control establishments in each year. Outcomes are measured in December of each year. Since the reform occurred in September 2012, December 2011 is the omitted reference year—the last period before treatment—while 2012 onward represents the post-treatment period.

The panel structure differs across outcomes. Wages and employment are observed annually in December, yielding a balanced panel with 2011 as the omitted reference year. CBA clause counts are observed only when an agreement is filed. Pre-treatment clause counts come from two agreements, the establishment’s earliest 2009 filing and its first renewal before 2012, the

latter serving as the omitted reference. We restrict post-treatment observations to CBAs filed starting in 2013, ensuring that post-treatment clause counts reflect negotiations conducted after the reform.²⁵

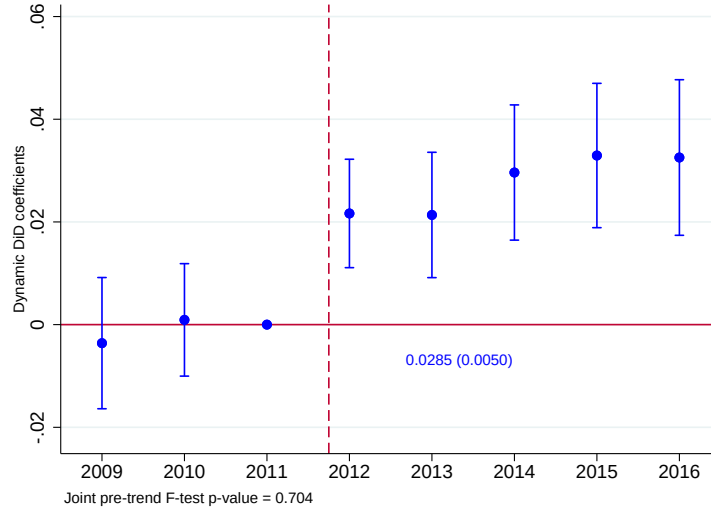
Identification. A causal interpretation of our estimates requires conditional parallel trends, that is, absent the reform, outcomes would have evolved similarly in directly treated and zero-connectivity untreated establishments, conditional on time-varying controls. We assess this assumption through placebo pre-trend tests, which we report for each outcome. Treatment status depends on whether a firm held an active CBA on September 25, 2012, which is a function of negotiation timing established before the reform. Because the reform and its announcement were unexpected (see Section 2), anticipation effects are unlikely. The observable differences between the two groups documented above concern levels, not trends, and so do not by themselves threaten this assumption. They could nonetheless predict differential outcome trajectories. Our time-varying controls guard against this, absorbing trend heterogeneity along the dimensions on which the groups differ most. The direct-effect wage estimates are further robust to controlling for finer bins and to workforce composition, as reported in Section 5.3.

Direct Effects Results. We find that higher collective bargaining power increased hourly wages at directly treated firms by 2.8% (Figure 2), and monthly wages by 2.6%. This was accompanied by an increase of 1.6 CBA clauses, a 5% increase compared to the pre-period mean. The effect on employment is not statistically significant (Table 1, Panel A).

When the comparison group instead includes all untreated establishments, the estimated effect on hourly wages is 2.0%, and 1.8% on monthly wages (Table 1, Panel B). Excluding connected firms from the comparison group thus increases the estimated wage effect by about 40%. Whether this increase is statistically distinguishable from zero cannot be assessed by comparing the two standard errors, because the specifications share the same directly treated establishments and use nested untreated groups, so the two estimates are not independent. We test their equality by stacking the samples from both panels into a single regression that fully interacts the difference-in-differences term with a Panel A vs. Panel B indicator, so that the interaction coeffi-

²⁵Because not all establishments file a CBA every year, clause count regressions have fewer observations. Since clause counts are indexed by agreement rather than by calendar year, these regressions also replace the year fixed effects in equation (3) with CBA-period fixed effects.

FIGURE 2: DIRECT EFFECTS ON HOURLY WAGES



Notes: This figure shows the event study estimates for the direct effects on log hourly wages of the ultractivity reform. It also reports the DiD estimate, that is, the estimate for β in the specification $Y_{it} = \alpha_t + \eta_i + \beta(D_i \times \mathbb{1}_{t \geq 2012}) + \sum_{k=2009}^{2016} \Gamma_k (X_i \times \mathbb{1}_{t=k}) + \varepsilon_{it}$. Standard errors are clustered at the establishment level. 95% confidence intervals are reported. The joint pre-trend test is of the null that the 2009 and 2010 coefficients are jointly zero. The dashed line indicates the timing of reform approval.

cient equals the difference between the two estimates and its standard error reflects their shared observations.²⁶ We reject equality of the direct wage effect across the two control-group definitions ($p = 0.012$ for hourly wages, $p = 0.009$ for monthly wages). The analogous differences for employment and the number of CBA clauses are small and statistically insignificant. Since the directly treated sample is identical across specifications, the attenuation when all untreated firms serve as the comparison group implies that connected untreated firms themselves experience positive wage spillovers.

5.2 Spillover Effects

We turn to estimating spillover effects explicitly, using connectivity to the treated as a continuous measure of indirect exposure. The analysis uses the full sample of untreated firms, regardless of their connectivity to directly treated establishments (Figure B2). Variation in C_i across these firms identifies the spillover effects.

²⁶The test is the cluster-robust Wald test of a cross-equation restriction in a stacked system (Wooldridge, 2010, Sec. 7.3), implemented following Weesie (1999).

TABLE 1: DIRECT EFFECTS OF UNION BARGAINING POWER

	Log Hourly Wages (1)	Log Monthly Wages (2)	Log Employment (3)	Clause Count (4)
Panel A: Treated vs. Zero-Connectivity Untreated				
Post $\times$ Treatment	0.0285*** (0.0050)	0.0262*** (0.0049)	−0.0013 (0.0152)	1.5544*** (0.3044)
Mean	2.648	7.767	3.320	32.316
Observations	112,695	112,695	113,112	73,181
Establishments	14,136	14,136	14,139	14,178
Pre-trend (placebo)	−0.0013 (0.0054)	−0.0002 (0.0046)	0.0156 (0.0101)	0.1465 (0.3484)
Panel B: Treated vs. All Untreated				
Post $\times$ Treatment	0.0202*** (0.0033)	0.0178*** (0.0032)	0.0024 (0.0111)	1.3481*** (0.2350)
Mean	2.613	7.743	3.484	31.586
Observations	130,708	130,708	131,232	84,080
Establishments	16,398	16,398	16,404	16,444
Pre-trend (placebo)	0.0048 (0.0035)	0.0042 (0.0030)	−0.0012 (0.0076)	0.2660 (0.2471)
Test: Panel A = Panel B				
<i>p</i> -value	[0.012]	[0.009]	[0.741]	[0.386]

Notes: This table presents difference-in-differences estimates of the ultractivity reform's direct effects (equation (3)). An establishment is directly treated if it had a firm-level CBA filed before September 25, 2012 with a contractual end date after that date. Panel A restricts the comparison group to untreated establishments with zero connectivity to directly treated firms; Panel B includes all untreated establishments. The final row reports the *p*-value of the test that the Post $\times$ Treatment effect is equal across the two panels. The test is estimated from a single stacked regression of the two samples in which each copy receives its own complete set of fixed effects and the difference-in-differences term is fully interacted with a copy indicator, so that the interaction coefficient equals the difference between the two panels. Standard errors are clustered at the establishment level, so that an establishment entering both copies forms a single cluster, which recovers the covariance between the two estimators and yields a valid test despite the overlapping samples. Wages and employment are measured in December of each year. Clause counts are observed only when a CBA is filed, reducing the number of observations in column (4). Post $\times$ Treatment reports the average effect for 2012–2016, relative to the pre-reform period (2009–2011). Mean reports the average of the dependent variable over 2009–2011 in each column's estimation sample. Pre-trend (placebo) reports the coefficient from a placebo regression estimated on pre-treatment data only (2009–2010 relative to 2011). All specifications include establishment fixed effects and year fixed effects interacted with three-digit industry, microregion, and negotiation-month indicators and with quartile bins of pre-treatment firm size, per-worker flows, and the outcome. Clause-count regressions substitute CBA-period fixed effects for year fixed effects. Standard errors clustered at the establishment level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Specification. The spillover analysis follows the same DiD structure as equation (3), but replaces the binary treatment indicator D_i with the continuous connectivity measure C_i :

$$(4) \quad Y_{it} = \alpha_t + \eta_i + \sum_{\substack{k=2009 \\ k \neq 2011}}^{2016} \delta_k (C_i \times \mathbb{1}_{t=k}) + \sum_{\substack{k=2009 \\ k \neq 2011}}^{2016} \Gamma_k (X_i \times \mathbb{1}_{t=k}) + \varepsilon_{it}$$

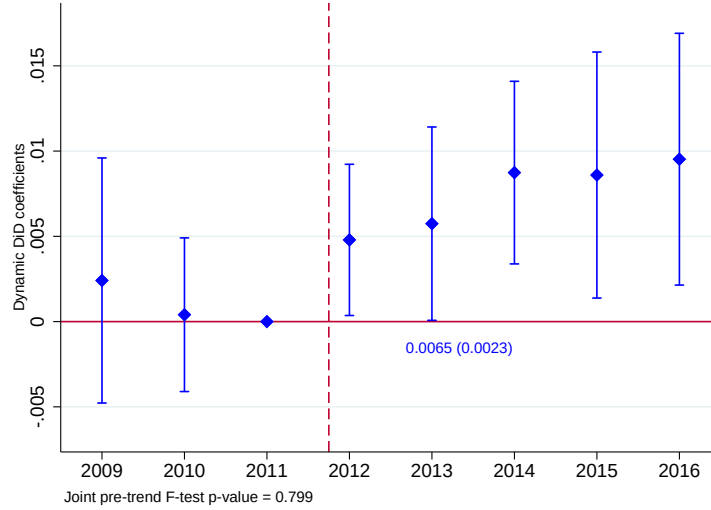
Our parameters of interest are δ_k , which measure the spillover effects for each year. We normalize C_i so that a value of 1 corresponds to the 90th percentile of connectivity across untreated firms ($\sim 3\%$ of the firm’s workforce), those competing most closely with the directly treated. Estimated δ_k thus represent the effect of moving from zero connectivity to the 90th percentile of C_i in the sample. We include the same set of time-varying controls as in the direct effects specification.

Identification. Our spillover estimates require that, absent the reform, the evolution of outcomes would have been uncorrelated with connectivity to treated firms, conditional on controls. A concern specific to the continuous design is that C_i is correlated with firm characteristics, namely establishment size and worker mobility, since large and high-turnover firms exchange more workers with all employers, including treated ones. The connectivity coefficient could then capture how these characteristics relate to outcomes, whether through differential trends or differential responses to contemporaneous conditions, rather than exposure to directly treated establishments. Our specification includes time-varying controls for firm size and per-worker-flow quartile bins, which absorb outcome variation associated with these characteristics in every period. We assess the resulting parallel-trends assumption directly through the placebo pre-trend tests, which are flat throughout. Section 5.3 provides evidence against the estimates being driven by firm characteristics or by local industry shocks. It further shows, through a more direct approach based on Borusyak and Hull (2023), that spillovers reflect connectivity to the directly treated rather than connectivity to comparable sets of firms.

Spillover Results. We find that firms highly connected to the directly treated raise hourly wages by 0.65% (0.50% for monthly wages) relative to firms with zero connectivity, as shown in Figure 3 and Table 2. Relative to direct effects of 2.8%, the spillover at the 90th percentile is roughly a quarter as large, a ratio of 0.23 (0.19 for monthly wages). We detect no spillover effects

on employment or on collectively bargained amenities, as measured by CBA clause counts (Table 2, Column 4). Although the reform raised both wages and amenities at directly treated establishments, the response at connected firms appears in wages alone, indicating that competitive pressure from improved outside options manifests in compensation primarily through wages. We further detail spillover amenity responses in Section 5.4.

FIGURE 3: SPILLOVER EFFECTS ON HOURLY WAGES



Notes: This figure plots the dynamic difference-in-differences coefficients δ_k from equation (4), with log hourly wage as the outcome. The sample consists of all untreated establishments. Connectivity C_i is normalized so that a value of 1 corresponds to the 90th percentile among untreated establishments; 2011 is the omitted reference year. The reported coefficient and standard error correspond to the pooled specification, which replaces the year-specific interactions with $C_i \times \mathbb{1}_{t \geq 2012}$. The joint pre-trend test is of the null that the 2009 and 2010 coefficients are jointly zero. The dashed line marks the approval of the reform (September 2012). Standard errors are clustered at the establishment level; bars represent 95% confidence intervals.

5.3 Robustness

A central concern for our empirical approach is that our estimates reflect firm characteristics correlated with connectivity rather than competition for workers. Flat pre-treatment coefficients weigh against such characteristics driving a general differential trend, but they could still shape how employers respond once the reform takes effect, and thus explain the wage estimates. Connectivity shows no significant relationship with pre-reform wages (Figure B3). Its relationship with employment attenuates once we add the industry, negotiation-month, and microregion controls included in our specification, while its relationship with clause counts persists. These linear fits leave the concern open, particularly for non-linear relationships they do not capture.

While our specification already bins pre-treatment size, per-worker flows, and the outcome

TABLE 2: SPILLOVER EFFECTS OF UNION BARGAINING POWER

	Log Hourly Wages (1)	Log Monthly Wages (2)	Log Employment (3)	Clause Count (4)
Post $\times$ Connectivity	0.0065*** (0.0023)	0.0050** (0.0022)	0.0009 (0.0081)	0.0227 (0.1175)
Mean	2.354	7.551	3.547	27.935
Observations	32,495	32,495	32,704	19,693
Establishments	4,084	4,084	4,088	4,142
Pre-trend (placebo)	0.0014 (0.0026)	0.0018 (0.0025)	−0.0015 (0.0063)	0.3256 (0.2014)

Notes: This table presents difference-in-differences estimates of the reform’s spillover effects (equation (4)) on the full sample of untreated establishments. Connectivity is normalized so that a value of 1 corresponds to the 90th percentile among untreated establishments. Wages and employment are measured in December of each year; clause counts are observed only when a CBA is filed, reducing the number of observations in column (4). Post $\times$ Connectivity reports the average effect for 2012–2016, relative to the pre-reform period (2009–2011). Pre-trend (placebo) reports the coefficient from a placebo regression estimated on pre-treatment data only (2009–2010 relative to 2011). Mean reports the average of the dependent variable over 2009–2011 in each column’s estimation sample. All specifications include establishment fixed effects and year fixed effects interacted with three-digit industry, microregion, and negotiation-month indicators and with quartile bins of pre-treatment firm size, per-worker flows, and the outcome. Clause-count regressions substitute CBA-period fixed effects for year fixed effects. Standard errors clustered at the establishment level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

into quartiles interacted with year fixed effects, we test whether absorbing differences in these characteristics with finer bins affects the results. Columns (2) and (3) of Table A2 show that the direct and spillover wage estimates remain stable under ten and twenty bins. Column (4) shows that the estimates are also not driven by differences in workforce characteristics, controlling for gender, race, education, age, and tenure.

Connectivity to the treated, through both observable and unobservable characteristics, may capture a wage response that a firm’s connectivity to any comparable set of employers would produce, rather than one specific to firms directly affected by the reform. The prior checks guard against this by absorbing observable characteristics. We also approach the confound directly, following the logic of Borusyak and Hull (2023). For each untreated firm we compute a counterfactual connectivity. We reshuffle the directly treated label within industry-by-negotiation-month cells, along which the likelihood of direct treatment varies (Figure B1), and average over 1,000 draws to obtain counterfactual connectivity μ_i .²⁷ Adding $\mu_i \times \text{Post}$ to the spillover specification lets

²⁷Controlling for $\mu_i \times \text{Post}$ is the control-function counterpart of recentring realized connectivity on its counterfactual mean. In natural experiments, where the assignment process is posited rather than known, Borusyak and Hull (2023) recommends this control over recentring, because it corrects the estimate when the process is right and leaves it unchanged when no confounding is present. Bergeaud et al. (2025) applies the procedure to an analogous spillover design, reading a null coefficient on counterfactual exposure as evidence that the estimated effect reflects true exposure to the treatment.

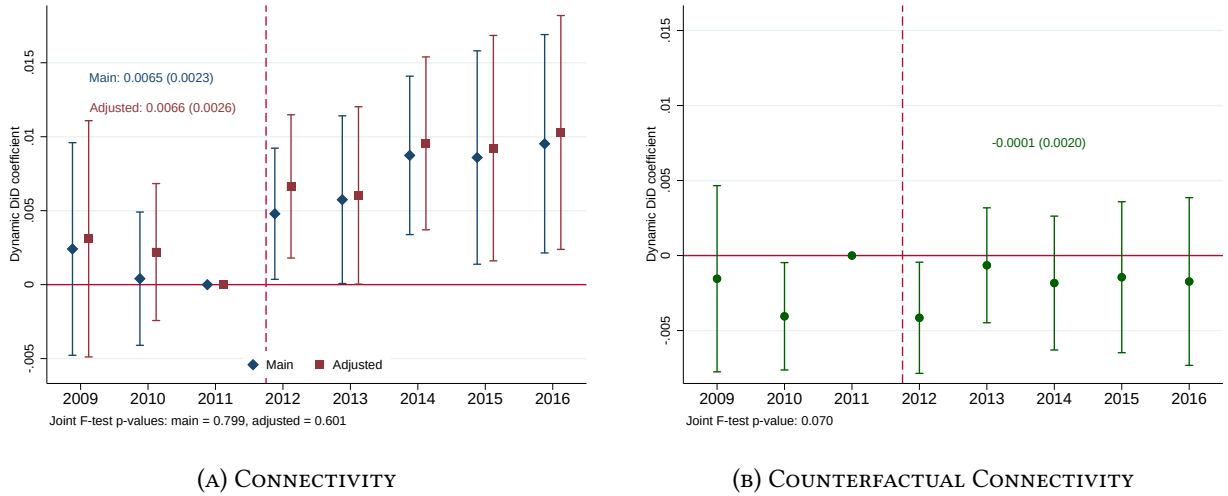
the identifying variation better isolate a firm's ties to the establishments actually treated rather than general connectedness. The spillover coefficient is largely unchanged, while counterfactual connectivity $\mu_i \times \text{Post}$ shows a small and insignificant coefficient (Figure 4).

A separate question is whether firm-specific connectivity captures competition beyond coarser definitions of local markets. Sharing an industry–microregion cell predicts connectivity (Figure 1, Panel A), so untreated firms in heavily treated cells are both more connected and more exposed to any common shock to the cell. In columns (5) and (6) of Table A2, we show effects are robust to controls for the share of firms or of employment directly treated. This evidence aligns with findings in Section 4 that connectivity is not reducible to shared industry–microregion classifications. Across all of our robustness checks, spillover and direct effects are stable, with the ratio between them ranging from 0.20 to 0.26.

5.4 Union Mediation of Spillovers

We next assess whether our spillover estimates reflect union activity rather than labor market competition. Separating the two is difficult in most settings approached by the literature, particu-

FIGURE 4: HOURLY WAGE SPILLOVERS: ROBUSTNESS TO COUNTERFACTUAL CONNECTIVITY



Notes: Event-study estimates of the hourly wage spillover on untreated establishments. Panel (a) plots the coefficients on connectivity to treated firms $\times$ year, from the baseline specification (*Main*) and from the specification that adds counterfactual connectivity $\times$ year as a control variable (*Adjusted*). Panel (b) plots the coefficients on counterfactual connectivity $\times$ year. Following the logic of Borusyak and Hull (2023), counterfactual connectivity is each untreated establishment's average connectivity to the treated set over 1,000 reshuffles of the treated label within industry $\times$ negotiation-month cells, holding the number treated per cell fixed and recomputing connectivity from the same pre-reform worker flows. In-plot text reports the pooled connectivity $\times$ Post estimate in each specification (panel a) and the pooled counterfactual $\times$ Post estimate (panel b), with standard errors in parentheses. Reported joint *F*-test *p*-values are for the null that the 2009 and 2010 coefficients are jointly zero. Bars are 95% confidence intervals from standard errors clustered at the establishment level.

larly the U.S.: because forming a union requires worker organization, wage gains at non-covered firms can reflect employers preempting the threat of unionization rather than competition for workers (Freeman and Medoff, 1981; Lewis, 1986; Fortin, Lemieux and Lloyd, 2021). Our setting does not present this confound, as union representation is universal and administratively assigned (Section 2).

The union landscape in Brazil, however, allows for potential union-mediated channels of its own. Because treatment assignment operates at the firm-union pair level, directly treated and untreated firms often share union coverage. The reform could therefore affect union behavior toward untreated firms. First, it could expand the union’s total bargaining resources, since union revenue is derived in part from mandatory contributions calculated as a fixed share of wages, which the reform raised at directly treated establishments.^{28,29} Second, it could alter the relative returns to effort across establishments. These channels need not move spillover outcomes in the same direction, leaving the net effect on untreated establishments ambiguous.³⁰ Either could confound our estimates, since worker flows between establishments correlate with common union coverage (Figure 1) and 91% of untreated establishments share a union with at least one directly treated firm. Union mediation, however, would operate through what unions actively produce, their negotiated provisions. If instead employers are responding to workers’ improved outside options, wages can rise with no change in bargained content.

Bargained content at connected firms does not respond on any margin we measure. While Table 2 shows no spillover effect on the total number of clauses, Table 3 shows this null does not mask offsetting movements across the three broad clause types, as counts and shares of wage-related, employment-related, and other clauses are all unchanged (columns (1) through (6)). The absence of movement in wage-related clauses is particularly relevant, as bargained wage provisions are where union-mediated transmission of the wage spillovers would most naturally appear.

Counts of types of clauses, however, weight every clause equally and thus may not capture how much workers value an agreement. Column (7) instead weights each clause by its wage-equivalent value, the log-point wage increase that would leave workers indifferent to the clause’s

²⁸Derenoncourt et al. (2025) documents a positive link between union revenues and wage premia in Brazil.

²⁹These mandatory contributions became optional following the 2017 Labor Reform.

³⁰Expanded capacity increases total effort available across all establishments, but the direction of reallocation depends on how the reform changes the marginal return to effort at directly treated relative to untreated firms. If ultractivity makes effort at directly treated firms sufficiently more productive, the union concentrates resources there, potentially offsetting or exceeding the capacity gain at untreated firms.

removal, and sums these weighted counts to express the agreement’s value in log wage points. We take the weights from Lagos (2026), estimated in the same setting.³¹ The estimate is close to zero and precise enough to rule out gains above 0.2%, roughly a third of the wage spillover.

TABLE 3: SPILLOVER EFFECTS ON CBA COMPOSITION AND VALUE

	Clause-type counts			Clause-type shares			Wage-equivalent value
	Wage (1)	Employment (2)	Other (3)	Wage (4)	Employment (5)	Other (6)	(7)
Post $\times$ Connectivity	−0.0180 (0.0276)	0.0581 (0.0380)	−0.0089 (0.0809)	−0.0018 (0.0013)	−0.0003 (0.0012)	0.0016 (0.0017)	−0.0009 (0.0016)
Mean	3.530	5.264	19.140	0.120	0.133	0.747	0.133
Observations	19,693	19,693	19,693	19,692	19,692	19,692	19,693
Establishments	4,142	4,142	4,142	4,142	4,142	4,142	4,142
Pre-trend (placebo)	0.0723 (0.0472)	0.0444 (0.0345)	0.2268* (0.1323)	0.0002 (0.0011)	0.0015 (0.0010)	−0.0018 (0.0013)	0.0019 (0.0029)

Notes: This table presents difference-in-differences estimates of the reform’s spillover effects (equation (4)) on the content of collective agreements, on the full sample of untreated establishments. Columns (1)–(3) report effects on the counts of wage-related, employment-related, and other clauses, following *Sistema Mediator*’s classification; columns (4)–(6) report effects on each type’s share of total clauses. Column (7) reports the effect on agreement value, the sum across the 24 *Sistema Mediator* subgroups of each subgroup’s clause count weighted by its wage-equivalent value from Lagos (2026), expressed in log wage points. Connectivity is normalized so that a value of 1 corresponds to the 90th percentile among untreated establishments. Post $\times$ Connectivity reports the average effect for 2012–2016, relative to the pre-reform period (2009–2011). Pre-trend (placebo) reports the coefficient from a placebo regression estimated on pre-treatment data only (2009–2010 relative to 2011). Mean reports the average of the dependent variable over 2009–2011 in each column’s estimation sample. All specifications include establishment fixed effects and CBA-period fixed effects interacted with three-digit industry, microregion, and negotiation-month indicators and with quartile bins of pre-treatment firm size, per-worker flows, and the outcome. Standard errors clustered at the establishment level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

As tests of union mediation, such content measures may be coarse. A union response could take forms these measures do not register, such as changes to specific provisions within the classifications, and still influence wages. We therefore turn to the wage effects, where any union channel behind the wage spillovers must register, and test whether estimates are sensitive to union exposure to the reform. We control for the share of a union’s covered establishments that are directly treated, and separately for the corresponding share of covered workers, including each control through either quartile or decile bins interacted with year. Across all four specifications the wage estimate is unchanged (Table A3).³²

³¹We take the per-subgroup weights directly from Table 6 of Lagos (2026), estimated in the same institutional setting from a revealed-preference regression relating workers’ ranking of establishments, inferred from job-to-job flows, to establishments’ wage premiums and clause counts. Agreement value is the sum over the 24 *Sistema Mediator* subgroups of each subgroup’s clause count times its weight, with subgroups not significantly valued receiving zero weight.

³²These controls absorb any union response that scales with the union’s own exposure to the reform. A within-union reallocation of effort toward or away from more-connected establishments would instead generate establishment-specific negotiated gains, of which the CBA content results show no evidence.

Taken together, these findings support the interpretation that wage gains at connected firms reflect competitive responses to workers' improved outside options rather than union-mediated transmission.

6 How Do Firms Adjust?

Following evidence that spillover effects operate through competition for workers, we now examine how firms adjust to and shape such competition. We first test whether the null employment effects mask adjustment along other margins, and whether firms shift the composition of their workforce. We then exploit variation in the types of workers who define the job mobility network to study responses across and within firms.

6.1 Employment and Workforce Composition

Employment Decomposition. While we detect no employment effect, employment policies can adjust on other margins. Employers facing a higher hourly cost of labor may contain the wage bill by cutting contracted hours, and employment can hold steady while hiring and separations move in the same direction. Table A4 reports estimates for contracted hours, retention, hiring, separation, quit, and layoff rates. No margin moves substantially.³³ At connected firms, workers' outside options improve without separations rising, and wages increase instead. This pattern is consistent with a wage response large enough to retain workers who would otherwise have left.

Workforce Composition. Higher wages at directly treated and connected firms can arise on two margins. Firms may pay more for a worker of a given type, or they may employ a different mix of workers. Only the first is a change in the price of labor, the margin the outside-option mechanism speaks to and the one that raises what a given worker earns. Which margin accounts for the wage effects is not clear a priori, and composition can work in either direction: a firm facing stronger competition may contain its wage bill by employing a cheaper workforce, or higher pay may attract and retain workers who command larger wages in the broader labor

³³Two coefficients warrant caution. The hiring rate at directly treated firms and the quit rate at highly connected firms display significant pre-trends. Both outcomes have post-treatment effects small relative to their pre-treatment means, at 4 and 2%, respectively, and neither carries the conclusions of this section.

market.

We separate the two margins by re-estimating the wage specifications using a residualized wage outcome, following a similar approach to Jäger, Schoefer and Heining (2021) and Friedrich (2022). We first estimate a worker-level Mincer regression of log hourly wages on a flexible and saturated set of worker characteristics, and then compute, for each establishment-year, the employer’s average residual.³⁴ We estimate the Mincer regression on the universe of RAIS employment records rather than on the analysis sample, so that the return to each characteristic approximates that of the overall formal labor market. Each residual then measures a worker’s pay relative to that of observationally equivalent workers nationwide, and the establishment-year outcome nets out the component of average wages predicted by workforce composition. This approach complements analyses of individual workforce characteristics, which we report in Table A5, since the residual weights each characteristic by its wage return in the labor market and allows that return to vary with workers’ other characteristics.

We find that firm-level wage effects are mostly price effects. At competing firms, residualized wages rise by 0.47% (Table 4), roughly three-quarters of the 0.65% raw effect. Most of the wage spillover therefore reflects higher pay for comparable workers. The remaining gap suggests that wage increases made connected firms more attractive to individuals with higher earning potential. At directly treated establishments, residualized wages rise by 2.89% against 2.85% for raw wages, so the direct effect operates on the price margin alone.

6.2 Outside Options Within the Firm

The spillover effects documented across firms could apply within them. Workers at the same employer may differ in the value of their outside options (Sharma, 2023; Caldwell and Danieli, 2024), and firms may respond by tailoring wages to this heterogeneity. Because our connectivity measure is built from worker-level job mobility flows, it can be computed separately for groups of workers within the same firm, allowing us to study these dynamics.

We partition workers three ways: by education (without vs. with a high school degree), by gender, and by tenure at the employer (fewer than twelve months vs. twelve months or more),

³⁴The Mincer regression projects log hourly wages on a quartic in age fully interacted with race $\times$ education $\times$ gender $\times$ year cells, with education measured at the full eleven-category resolution recorded in RAIS. We retain, for each worker and year, the employment relationships active throughout December, and select a single spell per worker-establishment pair.

TABLE 4: EFFECTS ON HOURLY WAGES RESIDUALIZED BY WORKFORCE CHARACTERISTICS

	Direct Effects		Spillover Effects	
	Raw (1)	Residualized (2)	Raw (3)	Residualized (4)
Post $\times$ Treatment	0.0285*** (0.0050)	0.0289*** (0.0054)		
Post $\times$ Connectivity			0.0065*** (0.0023)	0.0047** (0.0022)
Observations	112,695	112,695	32,495	32,495
Establishments	14,136	14,136	4,084	4,084
Pre-trend (placebo)	-0.0013 (0.0054)	-0.0040 (0.0057)	0.0014 (0.0026)	-0.0027 (0.0027)

Notes: This table presents difference-in-differences estimates of the ultractivity reform's effects on establishment-level log hourly wages. Columns (1)–(2) report direct effects, comparing directly treated establishments to untreated establishments with zero pre-reform connectivity. Columns (3)–(4) report spillover effects on the full sample of untreated establishments, using continuous connectivity to treated firms as the exposure measure, normalized so that a value of 1 corresponds to the 90th percentile. Columns (1) and (3) use raw log hourly wages, reproducing Table 1, column (1), and Table 2, column (1). Columns (2) and (4) use a Mincer-residualized log hourly wage. Worker-level residuals are obtained by projecting log hourly wages on a quartic age polynomial, with the intercept and all age coefficients free to vary across race $\times$ education $\times$ gender $\times$ year cells and education measured using all eleven categories recorded in RAIS. The Mincer regression is estimated on a broader sample of RAIS employment records over 2009–2016 (78.3 million workers across 5.8 million establishments), retaining employment relationships active throughout December and one spell per worker-establishment pair; cells with fewer worker-spell observations than regressors are dropped. Each residual thus measures a worker's pay relative to the national age-earnings profile of observationally equivalent workers, and the establishment-year outcome is the average of these residuals. Post $\times$ Treatment and Post $\times$ Connectivity measure average effects for 2012–2016. Pre-trend (placebo) reports the coefficient from a placebo regression estimated on pre-treatment data only (2009–2010 relative to 2011). All regressions include establishment fixed effects and year fixed effects interacted with three-digit industry, microregion, and negotiation-month indicators, as well as quartile-bin controls for pre-treatment establishment size, per-worker flows, and the outcome interacted with year fixed effects. Standard errors clustered at the establishment level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

and construct group-firm-specific connectivity measures analogous to equation (2).^{35,36} This finer measure may be estimated less precisely than its firm-level counterpart, and the partitions need not isolate the margins along which workers compete most directly. The groups nonetheless differ substantially in pay, most firms in the spillover sample employ both groups under every partition, and a considerable share of the variation in group connectivity lies across groups within the same firm rather than across firms (Table A6).

Wages and outside options within the firm. We first ask whether, within a given firm, the group more exposed to directly treated establishments experiences larger wage gains than its coworkers. Table A7, Columns 2 and 5, reports estimates including firm $\times$ year fixed effects,

³⁵For each group g within firm i , we define group-specific connectivity as $C_{ig} = \frac{1}{4} \sum_{t=2007}^{2010} \frac{\text{Flows with treated}_{ig,t,t+1}}{\text{Avg. employment}_{ig,t,t+1}}$, where flows and employment are restricted to workers belonging to group g .

³⁶These characteristics need not partition workers into groups facing identical alternatives, but wage differences across groups are large (Table A6), so workers in different groups plausibly face distinct sets of credible alternatives.

so identification comes exclusively from differential connectivity across groups within the same firm. Across the three partitions the within-firm wage estimates are centered near zero (Column 2). We thus find no evidence that firms tailor relative pay to specific competitive pressure, in line with theory and evidence on wage setting that emphasize company-wide pay policies and compressed wage differentials within firms (Manning, 1994; Giupponi and Machin, 2024; Amior and San, 2026).

This finding carries an implication for the estimation of labor market spillovers. A natural intuition is that exploiting more granular variation in competitive pressure should sharpen spillover estimates: if exposure varies meaningfully below the firm, measuring it at that finer level should use more of the available identifying variation. Our results suggest this intuition can mislead when the wage-setting unit faces constraints in differentiating pay across its workers. Column 3 makes this concrete. Relating each group’s wages to its own connectivity, but without firm $\times$ year fixed effects, identifies the coefficient from both across- and within-firm variation. The estimate therefore combines the significant across-firm response of Column 1 with the muted within-firm response of Column 2, leaving it below the firm-level baseline in every partition. This points to the importance of matching the level of measurement to the level of the wage-setting unit to recover labor market spillovers.

What type of competition drives the firm-level response? While we do not find that firms differentiate pay within their workforce, the firm-level response may still depend on the incidence of competition across worker types. We examine this by including both group-specific exposure measures from a given partition simultaneously, so that each coefficient approximates the contribution of competition originating among that group of workers, while outcomes remain at the firm level.

Our evidence points to spillovers operating through a firm’s longer-tenured workers (Table A8). Exposure running through workers with at least a year at the establishment raises wages, whereas exposure through shorter-tenured workers does not, and we reject equality of the two coefficients ($p = 0.010$). Under the education and gender partitions, the two groups’ exposures are statistically indistinguishable from one another. The pattern is consistent with higher replacement costs of longer-tenured workers prompting firms to increase wages, possibly driven by the prospect of losing firm-specific capital and a revealed-good match that an outside hire is

not expected to replace (Jäger, Heining and Lazarus, 2024).

7 Conclusion

We document wage spillovers from union bargaining power, where effects propagate through the outside options of workers. Using workers' transitions between employers to measure competition between firms, we show that workers more likely to perceive directly affected firms as credible alternative employers experience larger wage increases. These spillovers require no mediation from the institution generating the initial shock. Such a pattern suggests that for a labor market feature to generate spillovers, producing direct effects can be sufficient.

The mechanism may therefore operate across bargaining regimes that otherwise share little. Collective bargaining institutions differ markedly across countries in coverage, in how workers and firms are assigned to bargaining units, in the presence of automatic extensions, and in the degree of centralization (Jäger, Naidu and Schoefer, 2025). The channels through which unions are thought to reach the broader workforce are often tied to these architectures. Threat effects require that unionization be contestable (Fortin, Lemieux and Lloyd, 2021), and administrative extension requires legal machinery to bind firms that never negotiated (Hijzen and Martins, 2020; Martins, 2021). Competition for workers instead rests on the structure of the labor market rather than the design of the institution.

The same logic applies beyond collective bargaining. Any policy that affects compensation or employment at a subset of workplaces reshapes the outside options of workers elsewhere. Evidence from other settings is consistent with this, spanning wage floors (Demir, 2025), mergers (Arnold, 2025), privatization of state-owned firms (Arnold, 2022), and public-sector pay (Staiger, Spetz and Phibbs, 2010; Willén, 2021).

These spillovers bear on measurement. Standard comparisons between treated and untreated employers may place indirectly affected workplaces in the control group, biasing the resulting estimates. In our setting, spillovers run in the same direction as the direct effect, so estimates whose comparison group includes untreated firms connected through worker flows understate it. Accounting for spillovers may therefore be necessary not only to characterize the total impact of a policy but also to measure the partial effects that applied work often targets.

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Online Appendix

Outside Options and Collective Bargaining Spillovers

Luis de Azevedo Gomes

Guilherme Neri

August 4, 2026

A Appendix Tables

TABLE A1: SAMPLE DESCRIPTIVES

	Directly Treated	Untreated	
		All (Spillover Sample)	Zero-Connectivity
	(1)	(2)	(3)
<i>Panel A: Sample Size</i>			
Number of establishments	12,276	4,196	1,931
Establishment size	143.4	161.0	23.3
<i>Panel B: Workforce Characteristics</i>			
High School Degree (share)	0.73	0.67	0.68
Female (share)	0.37	0.37	0.41
Non-white (share)	0.31	0.34	0.32
Age (years)	36.9	35.4	36.1
Tenure (years)	6.8	4.8	5.3
<i>Panel C: Compensation</i>			
Mean Monthly Wage	3,624	2,748	2,596
Median Monthly Wage	3,112	2,327	2,296
Mean CBA Clause Count	31.9	27.1	28.2
<i>Panel D: Worker Flows</i>			
Per-worker flows	0.271	0.241	0.228
Network size	47.6	57.1	9.7

Notes: This table presents establishment-level descriptive statistics, averaged over 2009–2011 unless otherwise noted. Column (1) reports directly treated establishments; column (2) all untreated establishments; column (3) the subset of untreated establishments with zero connectivity to the directly treated ($C_i = 0$, equation (2)), which serve as the comparison group for the direct-effect estimates. Age and tenure are in years. Wages are December earnings in 2015 Brazilian Reais. Clause count is the number of provisions in the establishment’s first post-2009 CBA. Per-worker flows are total worker flows divided by average employment. Network size is the total number of distinct establishments with which the establishment saw workers transitioning. Per-worker flows and network size are measured over 2007–2011, matching the connectivity measurement window.

TABLE A2: ROBUSTNESS OF HOURLY WAGE EFFECTS

		Controls: # Bins		Controls:	Controls: Local Industry	
	Main	10 Bins	20 Bins	Workforce Characteristics	% Firms Treated	% Workers Treated
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Direct Effects						
Post $\times$ Treatment	0.0285*** (0.0050)	0.0283*** (0.0050)	0.0277*** (0.0050)	0.0284*** (0.0050)	0.0262*** (0.0059)	0.0298*** (0.0059)
Observations	112,695	112,695	112,695	112,695	112,695	112,695
Establishments	14,136	14,136	14,136	14,136	14,136	14,136
Pre-trend (placebo)	-0.0013 (0.0054)	-0.0010 (0.0055)	-0.0011 (0.0055)	-0.0014 (0.0054)	0.0027 (0.0059)	0.0098 (0.0061)
Panel B: Spillover Effects						
Post $\times$ Connectivity	0.0065*** (0.0023)	0.0073*** (0.0024)	0.0070*** (0.0024)	0.0066*** (0.0023)	0.0062*** (0.0023)	0.0061*** (0.0023)
Spillover / direct effect	0.23	0.26	0.25	0.23	0.24	0.20
Observations	32,495	32,495	32,495	32,495	32,495	32,495
Establishments	4,084	4,084	4,084	4,084	4,084	4,084
Pre-trend (placebo)	0.0014 (0.0026)	0.0012 (0.0026)	0.0009 (0.0026)	0.0013 (0.0026)	0.0013 (0.0026)	0.0013 (0.0026)

Notes: This table summarizes the robustness of the main log hourly wage results across alternative specifications. Panel A reports direct effects using zero-connectivity untreated establishments as the comparison group; Panel B reports spillover effects on the full sample of untreated establishments. The reported ratio divides the Panel B estimate by the Panel A estimate from the same column. Column (1) is the baseline specification with quartile-bin controls. Columns (2)–(3) replace quartile bins with N -bin controls for pre-treatment firm size, per-worker flows, and hourly wages, interacted with year fixed effects. Column (4) adds quartile-bin controls, interacted with year fixed effects, for pre-treatment workforce characteristics: gender composition (% male), race composition (% white), education (% with at least a high school degree), average age, and average tenure. Columns (5)–(6) add, interacted with year fixed effects, quartile bins of the share of directly treated firms (column 5) or of employment in directly treated firms (column 6) within each establishment's industry $\times$ microregion cell. Standard errors clustered at the establishment level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE A3: ROBUSTNESS OF HOURLY WAGE SPILLOVERS TO UNION CONTROLS

	Main (1)	% of union's firms treated		% of union's workers treated	
		Quartiles (2)	Deciles (3)	Quartiles (4)	Deciles (5)
Post $\times$ Connectivity	0.0065*** (0.0023)	0.0067*** (0.0023)	0.0071*** (0.0023)	0.0067*** (0.0023)	0.0065*** (0.0023)
Spillover / direct effect	0.23	0.24	0.25	0.24	0.23
Observations	32,495	32,495	32,495	32,495	32,495
Establishments	4,084	4,084	4,084	4,084	4,084
Pre-trend (placebo)	0.0014 (0.0026)	0.0008 (0.0026)	0.0007 (0.0025)	0.0009 (0.0025)	0.0005 (0.0026)

Notes: This table assesses the robustness of the spillover hourly wage estimates to controls for union exposure to the reform, on the full sample of untreated establishments. The reported ratio divides each column's spillover estimate by the main direct-effect estimate (0.0285). Column (1) reproduces the baseline specification (equation (4)), which already includes year-interacted quartile bins of pre-treatment firm size, per-worker flows, and the outcome. Columns (2)–(5) add to that baseline a control for the share of the establishment's union that is directly treated, interacted with year fixed effects. The share is measured over the union's covered establishments in columns (2)–(3) and over its covered workers in columns (4)–(5), and enters as quartile indicators in columns (2) and (4) and decile indicators in columns (3) and (5). Standard errors clustered at the establishment level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE A4: DECOMPOSING EMPLOYMENT EFFECTS

	Log Hours (1)	Retention Rate (2)	Hiring Rate (3)	Separation Rate (4)	Quit Rate (5)	Layoff Rate (6)
Panel A: Direct Effects						
Post $\times$ Treatment	0.0015 (0.0153)	0.0060* (0.0032)	0.0166 (0.0128)	−0.0047 (0.0125)	−0.0031 (0.0036)	0.0039 (0.0070)
Mean	6.983	0.816	0.426	0.375	0.092	0.175
Observations	113,112	112,620	113,112	113,112	113,112	113,112
Establishments	14,139	14,139	14,139	14,139	14,139	14,139
Pre-trend (placebo)	0.0127 (0.0105)	0.0011 (0.0056)	−0.0378** (0.0169)	−0.0149 (0.0159)	−0.0066 (0.0048)	−0.0120 (0.0091)
Panel B: Spillover Effects						
Post $\times$ Connectivity	0.0001 (0.0080)	0.0014 (0.0017)	−0.0096 (0.0095)	−0.0035 (0.0066)	−0.0027* (0.0015)	−0.0017 (0.0048)
Mean	7.284	0.763	0.579	0.517	0.118	0.262
Observations	32,704	32,471	32,704	32,704	32,704	32,704
Establishments	4,088	4,088	4,088	4,088	4,088	4,088
Pre-trend (placebo)	−0.0012 (0.0063)	0.0036 (0.0033)	−0.0079 (0.0220)	−0.0165 (0.0183)	−0.0072** (0.0034)	−0.0090 (0.0131)

Notes: This table presents difference-in-differences estimates of the reform's effects on establishment-level employment-related outcomes. Panel A reports direct effects, comparing directly treated establishments to untreated establishments with zero pre-reform connectivity. Panel B reports spillover effects on the full sample of untreated establishments, using continuous connectivity to treated firms as the exposure measure, normalized so that a value of 1 corresponds to the 90th percentile. Log hours is the log of total weekly contracted hours summed across December-employed workers. The retention rate is the share of January-employed workers who were still employed at the end of year t . It is undefined for a small number of establishment-years, which reduces its number of observations. The hiring, separation, quit, and layoff rates are the number of hires, total separations, voluntary separations, and employer-initiated dismissals during year t , respectively, each scaled by average employment between years $t - 1$ and t . Because separations are recorded as flows over the full year t , including workers hired and separated within the year, the separation rate need not equal one minus the retention rate. Total separations include voluntary departures, employer-initiated dismissals, retirements, and deaths. Mean reports the pre-treatment average (2009–2011) across establishments in each panel's estimation sample. Post $\times$ Treatment and Post $\times$ Connectivity measure average effects for 2012–2016. Pre-trend (placebo) reports the coefficient from a placebo regression estimated on pre-treatment data only (2009–2010 relative to 2011). All regressions include establishment fixed effects and year fixed effects interacted with three-digit industry, microregion, and negotiation-month indicators, as well as quartile-bin controls for pre-treatment establishment size, per-worker flows, and the outcome, interacted with year fixed effects. Standard errors clustered at the establishment level in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE A5: EFFECTS ON WORKFORCE COMPOSITION

	Avg. Age (1)	Male Share (2)	White Share (3)	High School+ Share (4)
Panel A: Direct Effects				
Post $\times$ Treatment	−0.0080 (0.0976)	−0.0091*** (0.0035)	−0.0061 (0.0048)	−0.0055 (0.0038)
Mean	36.804	0.621	0.686	0.716
Observations	113,112	113,112	113,112	113,112
Establishments	14,139	14,139	14,139	14,139
Pre-trend (placebo)	−0.1501 (0.0915)	0.0055* (0.0034)	−0.0028 (0.0046)	0.0034 (0.0037)
Panel B: Spillover Effects				
Post $\times$ Connectivity	−0.0117 (0.0443)	0.0003 (0.0019)	0.0004 (0.0022)	−0.0018 (0.0021)
Mean	35.417	0.632	0.667	0.664
Observations	32,704	32,704	32,704	32,704
Establishments	4,088	4,088	4,088	4,088
Pre-trend (placebo)	0.0510 (0.0403)	−0.0013 (0.0018)	−0.0019 (0.0020)	0.0013 (0.0015)

Notes: This table presents difference-in-differences estimates of the reform's effects on establishment-level workforce composition. Panel A reports direct effects, comparing directly treated establishments to untreated establishments with zero pre-reform connectivity. Panel B reports spillover effects on the full sample of untreated establishments, using continuous connectivity to treated firms as the exposure measure, normalized so that a value of 1 corresponds to the 90th percentile. Average age is the mean age of December-employed workers. High school+ is the share of workers with at least a completed high school degree. Mean reports the pre-treatment (2009–2011) average across establishments in each panel's estimation sample. Post $\times$ Treatment and Post $\times$ Connectivity measure average effects for 2012–2016. Pre-trend (placebo) reports the coefficient from a placebo regression estimated on pre-treatment data only (2009–2010 relative to 2011). All regressions include establishment fixed effects and year fixed effects interacted with three-digit industry, microregion, and negotiation-month indicators, as well as quartile-bin controls for pre-treatment establishment size, per-worker flows, and the outcome, interacted with year fixed effects. Standard errors clustered at the establishment level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE A6: GROUP-LEVEL CONNECTIVITY DESCRIPTIVES

	<i>Education</i>		<i>Gender</i>		<i>Tenure</i>	
	No high school	High school+	Female	Male	<12 months	≥12 months
Avg. employment	60.2	104.3	55.7	107.9	48.8	110.1
Avg. hourly wage	9.9	14.8	12.2	14.7	11.7	14.7
Avg. monthly wage	1,796	2,613	2,149	2,596	2,054	2,610
Avg. per-worker flows	0.2055	0.2732	0.2449	0.2687	0.5208	0.1083
Connectivity	0.0094	0.0148	0.0103	0.0149	0.0259	0.0067
Share of firms w/ both groups	0.86		0.88		0.97	
Share of connectivity variance within firms	0.27		0.34		0.45	
Share of connectivity variance between firms	0.73		0.66		0.55	

Notes: This table summarizes pre-treatment (2009–2011) descriptive statistics for group-level connectivity under three worker partitions. The unit of observation is a group $\times$ firm cell. Tenure groups are defined by whether a worker has completed twelve months at the establishment. “Avg. hourly wage” and “Avg. monthly wage” are group means of real December earnings, per contracted hour and per month, respectively, in 2015 Brazilian Reais. “Avg. per-worker flows” is total worker flows per group worker, averaged across year pairs 2007–08 through 2010–11. “Connectivity” is the average pre-treatment worker flow between a firm’s group- g workers and directly treated establishments, per group- g worker (footnote 35); a firm’s connectivity therefore varies across its worker groups. “Share of firms w/ both groups” is the share of spillover-sample firms with non-missing connectivity for both groups. The final two rows decompose the total variance of group connectivity into its within- and between-firm components on firms with both groups, following the ANOVA identity $\text{Var}(X) = E[\text{Var}(X \mid \text{firm})] + \text{Var}(E[X \mid \text{firm}])$, and report each as a share of the total.

TABLE A7: GROUP-LEVEL SPILLOVER EFFECTS: EDUCATION, GENDER, AND TENURE

	Log Hourly Wages			Log Employment		
	Firm-level (1)	Within firms (2)	Overall (3)	Firm-level (4)	Within firms (5)	Overall (6)
Panel A: Education (no HS / HS+)						
Connectivity $\times$ Post	0.0065*** (0.0023)			0.0009 (0.0081)		
Group-Connectivity $\times$ Post		-0.0039 (0.0043)	0.0025 (0.0023)		-0.0123 (0.0102)	-0.0074 (0.0068)
Observations	32,495	50,612	57,786	32,704	50,612	57,789
Groups $\times$ firms	—	6,938	7,564	—	6,938	7,565
Establishments	4,084	3,469	4,084	4,088	3,469	4,085
Pre-trend (placebo)	0.0014 (0.0026)	0.0054 (0.0057)	0.0012 (0.0029)	-0.0015 (0.0063)	0.0101 (0.0101)	0.0042 (0.0050)
Panel B: Gender (female / male)						
Connectivity $\times$ Post	0.0065*** (0.0023)			0.0009 (0.0081)		
Group-Connectivity $\times$ Post		0.0023 (0.0022)	0.0046** (0.0019)		-0.0006 (0.0047)	0.0036 (0.0048)
Observations	32,495	53,592	59,283	32,704	53,592	59,289
Groups $\times$ firms	—	7,156	7,670	—	7,156	7,672
Establishments	4,084	3,578	4,084	4,088	3,578	4,085
Pre-trend (placebo)	0.0014 (0.0026)	0.0002 (0.0053)	0.0002 (0.0024)	-0.0015 (0.0063)	-0.0067 (0.0110)	-0.0021 (0.0061)
Panel C: Tenure (<12mo / $\geq$12mo)						
Connectivity $\times$ Post	0.0065*** (0.0023)			0.0009 (0.0081)		
Group-Connectivity $\times$ Post		-0.0007 (0.0024)	0.0024 (0.0023)		-0.0038 (0.0067)	0.0021 (0.0059)
Observations	32,495	54,650	59,784	32,704	54,650	59,786
Groups $\times$ firms	—	7,846	8,041	—	7,846	8,042
Establishments	4,084	3,923	4,084	4,088	3,923	4,085
Pre-trend (placebo)	0.0014 (0.0026)	0.0036 (0.0038)	0.0048* (0.0026)	-0.0015 (0.0063)	-0.0108 (0.0089)	-0.0104 (0.0065)
Additional Controls						
Group-level Variables		✓	✓		✓	✓
Firm $\times$ Year FE		✓			✓	

Notes: Each panel partitions workers into two groups and estimates spillover effects at the firm and group level, on the sample of untreated, balanced-panel establishments. The partitions are by education (Panel A), gender (Panel B), and tenure (Panel C). The education partition separates workers without a high school degree from workers with at least a high school degree; the tenure partition separates workers with fewer than twelve months at the establishment from workers with twelve months or more. Group-Connectivity is defined analogously to firm-level Connectivity (equation (2)), restricting both the worker flows to directly treated establishments and the employment that normalizes them to workers in the group. Both measures are scaled so that a value of 1 corresponds to the 90th percentile of the firm-level distribution. Columns (1) and (4) report the full-sample firm-level spillover of Table 2, identical across panels. In the “Within firms” and “Overall” columns, the unit of observation is a group $\times$ firm $\times$ year cell and all regressions include group $\times$ firm fixed effects, together with group-interacted year paths for industry, microregion, and negotiation month. “Within firms” columns add firm $\times$ year fixed effects, so identification comes exclusively from differential connectivity across groups within the same firm-year; these absorb all firm-level controls. “Overall” columns instead add the firm-level pre-treatment wage, employment, and per-worker-flow bins interacted with year, the same controls the firm-level column uses, so identification combines within- and across-firm variation. Group-level controls are quartile bins of pre-treatment group wage, group employment, and group per-worker flows, each allowed to follow its own year path by worker group. Pre-trend (placebo) reports the coefficient from a placebo regression estimated on pre-treatment data only (2009–2010 relative to 2011). Standard errors clustered at the establishment level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

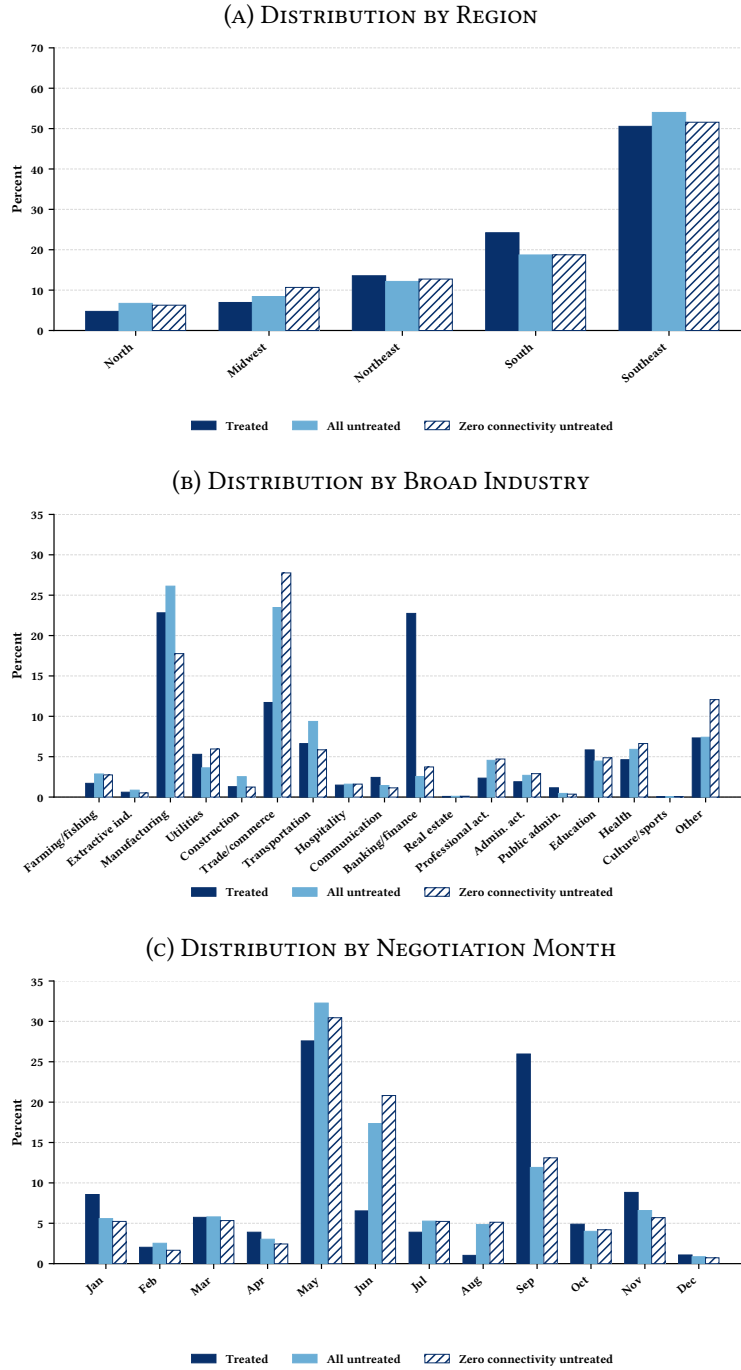
TABLE A8: GROUP-SPECIFIC CONNECTIVITY AND FIRM-LEVEL OUTCOMES

<i>Firm-level outcome:</i>	Post $\times$ Connectivity		Pre-trend (placebo)	
	Log Hourly Wages (1)	Log Employment (2)	Log Hourly Wages (3)	Log Employment (4)
Panel A: Education (no HS / HS+)				
Low-Education Connectivity	0.0027 (0.0021)	0.0027 (0.0077)	-0.0008 (0.0027)	0.0009 (0.0067)
High-Education Connectivity	0.0032 (0.0024)	0.0034 (0.0076)	0.0017 (0.0024)	0.0025 (0.0056)
Equality test (p -value)	0.873	0.951		
Observations	28,227	28,368		
Establishments	3,546	3,546		
Panel B: Gender (female / male)				
Female Connectivity	0.0019* (0.0011)	0.0023 (0.0031)	0.0006 (0.0016)	0.0011 (0.0025)
Male Connectivity	0.0026 (0.0017)	0.0020 (0.0051)	0.0017 (0.0027)	-0.0008 (0.0057)
Equality test (p -value)	0.752	0.956		
Observations	28,900	29,056		
Establishments	3,631	3,632		
Panel C: Tenure (<12mo / $\geq$12mo)				
Short-Tenure Connectivity	0.0005 (0.0013)	-0.0034 (0.0039)	0.0021 (0.0013)	-0.0006 (0.0034)
Long-Tenure Connectivity	0.0088*** (0.0028)	0.0093 (0.0085)	0.0002 (0.0024)	0.0034 (0.0056)
Equality test (p -value)	0.010	0.168		
Observations	31,829	32,000		
Establishments	4,000	4,000		

Notes: Each panel partitions workers into two groups and jointly estimates the spillover effects of the two resulting group-specific connectivity measures on firm-level outcomes, on the sample of untreated establishments. Low-education workers are those without a high school degree and high-education workers those with at least a high school degree; short-tenure workers are those with fewer than twelve months at the establishment and long-tenure workers those with twelve months or more. Both measures enter regressions simultaneously. Group-specific connectivity is defined as in Table A7, and both measures are scaled so that a value of 1 corresponds to the 90th percentile of the firm-level distribution. The unit of observation is a firm $\times$ year cell, and all regressions include the fixed effects and time-varying controls of equation (4). Columns (1)–(2) report average effects for 2012–2016, relative to the pre-reform period (2009–2011). Columns (3)–(4) report coefficients from placebo regressions estimated on pre-treatment data only (2009–2010 relative to 2011); the equality test, observation counts, and firm counts refer to the specifications in columns (1)–(2). Equality test reports the p -value of the test that the two coefficients in the column are equal, using the normal approximation. Samples are smaller than in Table 2 because both group-specific measures must be non-missing. Standard errors clustered at the establishment level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

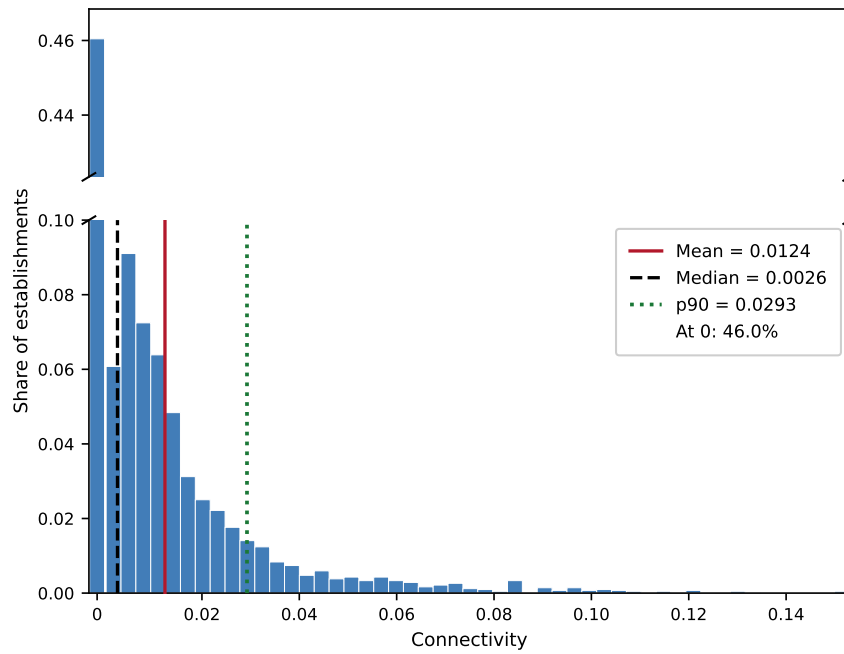
B Appendix Figures

FIGURE B1: DISTRIBUTION OF ESTABLISHMENTS BY CHARACTERISTICS ACROSS TREATMENT GROUPS



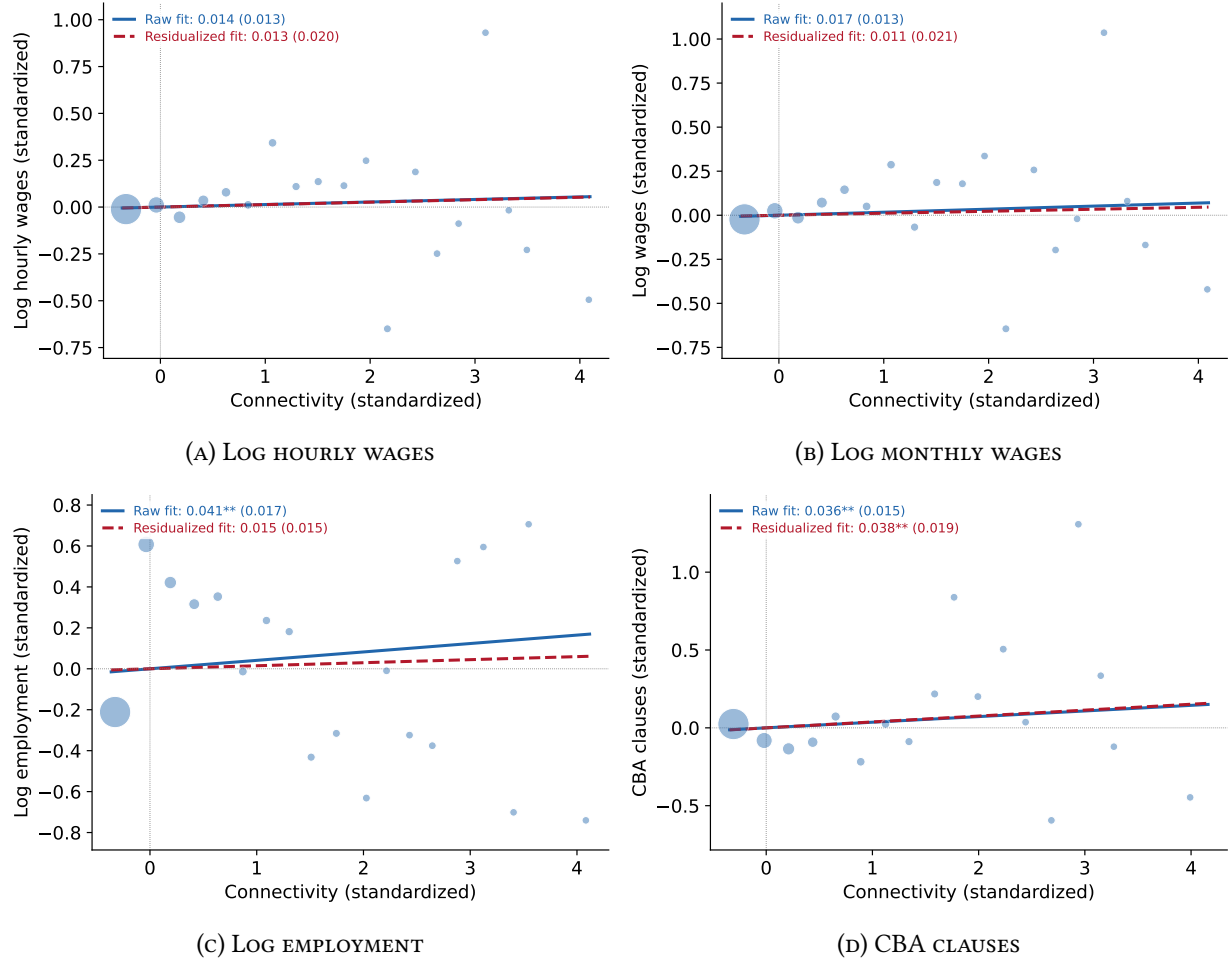
Notes: This figure shows the distribution of establishments across regions, broad industry groups, and CBA negotiation months, separately for directly treated establishments, all untreated establishments, and zero-connectivity untreated establishments.

FIGURE B2: DISTRIBUTION OF CONNECTIVITY TO DIRECTLY TREATED FIRMS



Notes: Histogram of connectivity to directly treated establishments across the spillover estimation sample of untreated establishments. Connectivity C_i , defined in equation (2), is the per-worker count of pre-reform (2007–2011) worker flows between the establishment and directly treated firms. The dashed line marks the median, the solid line the mean, and the dotted line the 90th percentile, each reported in the legend. The vertical axis is broken so that the zero-connectivity bar, which holds 46.0% of establishments, does not compress the remaining bars. The horizontal axis is truncated at the 99th percentile for readability.

FIGURE B3: MAIN OUTCOMES AND CONNECTIVITY



Notes: Binned scatterplots of each pre-treatment outcome against connectivity, across untreated establishments. Dots are bin means of the raw data. Both axes are standardized. The solid blue line is the raw OLS fit. The dashed red line is the OLS fit from the *residualized* regression, in which both the outcome and connectivity are partialled out on industry, negotiation-month, and microregion fixed effects, which are included in our specification. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C Data Appendix

C.1 Sample Construction

Our analysis combines two administrative sources. The first is *Relação Anual de Informações Sociais* (RAIS), the matched employer-employee register of formal employment in Brazil. The second is the registry of collective bargaining agreements (CBAs) filed in *Sistema Mediador*, kept by the Brazilian Ministry of Labor and assembled as a dataset by Lagos (2026).³⁷ Since 2009, the Ministry requires all CBAs to be registered in *Sistema Mediador* to acquire legal validity. Filing parties enter the signatories, the term and filing dates, the municipalities covered, and the text of each clause, classifying every clause into one of a predefined list of types. Because a firm-level agreement need not cover every establishment of a firm, we separately scraped each agreement’s municipalities of coverage from *Sistema Mediador* and use them to determine which establishments fall within an agreement’s scope. Municipalities are assigned to microregions using the crosswalk published by the Brazilian statistical agency (IBGE).

The unit of observation throughout is the establishment, identified by its fourteen-digit tax identifier (CNPJ). As in the main text, we use firm and establishment interchangeably, except when matching agreements to employers, where the distinction bears on the procedure and we note it explicitly.

We construct three samples: 1) an establishment-year panel of formal labor market outcomes drawn from RAIS, 2) an establishment-year record of collective bargaining drawn from the CBA registry, and 3) the analysis sample formed by merging the two and imposing restrictions. A fourth, separate sample of worker transitions is used to construct the connectivity measure. We describe each in turn.

C.1.1 The establishment-year panel from RAIS

Each annual filing reports one record per employment contract held at the establishment during the year, so a worker with several contracts appears more than once. We reduce these records to an establishment-year panel spanning 2009 to 2016. For each year, we measure employment

³⁷The CBA registry data can be accessed at <https://sites.google.com/view/lorenzolagos/data>.

and compensation in December, as records for wages in each month of the year are not available before 2015. We restrict to spells that were active on December 31st in every year, and in which the tenure was longer than one month. We impose no restriction on contract type.

A worker may hold several concurrent spells at the same establishment, so even after restricting to December spells, we may have more than one spell per worker in the same establishment. We further impose additional criteria sequentially to select a single spell per worker in a year. First, we choose the spell with the highest contracted hours. Second, we choose the one with the highest hourly wage. Any remaining ties are broken with a uniform random draw under a fixed seed. December employment at an establishment is the resulting headcount.

C.1.2 Collective bargaining coverage from the CBA registry

From the CBA registry (Lagos, 2026) we construct a record of the agreements for each establishment over the sample period, imposing the restrictions described below. We keep agreements flagged as valid in the registry and with non-missing start, end, and filing dates. We require the filing date to fall between the start and end dates of the contract term, which discards entries whose dates are internally inconsistent, and we drop agreements whose term ended before 2009, the first year of the registry’s electronic mandate. The interval between the start date and the filing date measures the length of the negotiation.

The registry distinguishes firm-level agreements (*acordos coletivos*), signed between a union and an individual employer, from sectoral agreements (*convenções coletivas*), signed between a union and an employer association on behalf of all firms in a category-geography cell. We retain firm-level agreements only.

We scraped additional information on the geographic coverage of each CBA from *Sistema Mediador*.³⁸ Agreements are recorded at three levels of geographic coverage: specific municipalities, entire states, and national coverage. Matching these agreements to employers is the one step where the firm-establishment distinction binds. Agreements are signed by the firm, identified by the first eight digits of the CNPJ, and headquarters commonly negotiate on behalf of subsidiaries located elsewhere. We therefore match on those eight digits interacted with the coverage geography, taking the municipal, state, and national tiers in turn. An establishment falls under an agreement when its firm signed the agreement and the establishment sits within the agreement’s

³⁸The platform is available at <https://mediador.trabalho.gov.br/sistemas/mediador>.

coverage area.

Although workers at an establishment are almost always represented by a single union, exceptions occur where industry-based and occupation-based categories overlap. We assign each establishment a main union, corresponding to the modal union across its agreements. Agreements negotiated by other unions are discarded, so that an establishment's bargaining history reflects a single counterpart.

We assign each agreement to every establishment-year from the year it takes effect through the last year in which it remains in force at year-end. Occasionally more than one agreement is on file for the same establishment and year, because an agreement was amended or re-filed, or because the union negotiated more than once. We combine these into a single record running from the earliest start date to the latest end date, counting each clause type once so that an amendment does not inflate the clause count. Each negotiation is dated by its filing date, averaged where several agreements are combined.

C.1.3 The analysis sample

We merge collective bargaining coverage with the RAIS establishment-year panel. To reach our analysis sample, we then impose two restrictions. The first concerns engagement in collective bargaining. We require each establishment to have filed an agreement in calendar 2009, at least one further agreement filed after that first 2009 agreement and before 1 January 2012, and at least one agreement filed after 2012. The first two conditions restrict the sample to bargaining units that negotiate frequently, so that untreated establishments are those that happened to be between agreements when the reform took effect rather than employers that rarely negotiate. Requiring two pre-reform negotiations also makes pre-trends in the content of agreements observable for every establishment in the sample. The third condition ensures that there is a post-reform negotiation whose content we can analyze.

The second restriction is to a balanced panel. We keep establishments present in all eight years from 2009 through 2016, so that estimates reflect changes within a fixed set of continuing establishments rather than shifts in sample composition across years. We end the panel in 2016 because a labor market reform passed in 2017 repealed the legal rule underlying our variation.

Wages and employment are observed every December, but the content of an agreement is

observed only when an agreement is filed, so clause outcomes are indexed by negotiation period rather than by calendar year. Period 1 is the establishment’s earliest 2009 agreement and period 2 its first renewal before 2012. Periods 3 through 6 are filings in calendar 2013, 2014, 2015, and 2016. Agreements filed during 2012 belong to no period and are excluded, so that every post-reform observation reflects a negotiation conducted after the reform took effect, and pre-reform observations reflect negotiations concluded before it. Regressions on clause outcomes replace the year fixed effect with the CBA-period fixed effect throughout, including in the time-varying controls, and use period 2 (2010–2011) as the omitted reference.

C.1.4 The worker flows sample

Connectivity is measured from worker movements between establishments over the four consecutive year pairs preceding the reform: 2007–08, 2008–09, 2009–10, and 2010–11. This sample is built separately from the establishment-year.

In each year we retain workers employed throughout December, as in Section C.1.1, and select one spell per worker-establishment pair by the same rule. We then assign each worker to a single establishment economy-wide, ranking on tenure (longest first), then on the real hourly wage (highest first), with a random tiebreak. A worker assigned to A in year t and to B in year $t + 1$ counts as a flow between the two establishments. Flows are undirected: the flow between A and B pools movers in each direction, with every worker contributing once.

C.1.5 Auxiliary samples

The residualized wage outcome requires returns to worker characteristics estimated on the broader labor market rather than on the analysis sample. We therefore estimate the Mincer regression on all RAIS employment records over 2009–2016, restricted to employment relationships active throughout December, with one spell per worker-establishment pair selected under the rule of Section C.1.1. The resulting sample covers 78.3 million workers across 5.8 million establishments. Cells with fewer worker-spell observations than regressors are dropped. Residuals are then averaged to the establishment-year over the analysis sample.

The group samples repeat the procedure of Section C.1.4 separately for each side of three worker partitions: education (without versus with a high school degree), gender, and tenure

(fewer than twelve months versus twelve months or more at the establishment). Within each partition, flows and employment are restricted to workers belonging to the group, so that group connectivity measures movements of that group’s workers per group worker. Workers are assigned to groups by their status in the origin year of each transition. An establishment contributes to a group only if it employs workers of that group, so the group samples are smaller than the firm-level sample.

C.2 Variable definitions

This appendix defines the variables used in the paper. Throughout, workers are those employed throughout the month of December and one spell is retained per worker-establishment pair, as described in Appendix C.1. Monetary values are adjusted to match December 2015 prices.

- **Wages.** RAIS records, for each spell, the earnings paid in December and the worker’s contracted weekly hours. We use two measures. Monthly wages are the December earnings. Hourly wages divide it by contracted weekly hours times 4.348, the mean number of weeks per month over the span of a year. Both measures are deflated and expressed in logs at the worker level, then averaged over the establishment’s December-employed workers.
- **Employment.** The count of distinct workers employed throughout December.
- **Hours.** Contracted weekly hours summed over the establishment’s December-employed workers.
- **Hiring rate.** Distinct workers hired at any point during year t , divided by average employment (across years $t - 1$ and t). A worker hired and separated within the same year still counts as a hire.
- **Separation rate.** Distinct workers who separated at any point during year t , divided by average employment. Separations include quits, layoffs, retirements, and deaths.
- **Quit rate.** Separations restricted to employee-initiated voluntary departures, divided by average employment.
- **Layoff rate.** Separations restricted to employer-initiated dismissals, divided by average employment.

- **Retention rate.** The share of workers employed throughout January of year t who are still at the establishment throughout the following December. January employment is defined symmetrically to December employment: hired before 1 January, not separated during January, and with more than one month of tenure as of 31 January. Because its denominator is January employment rather than average employment, retention is not one minus the separation rate. The rate is undefined where an establishment has no January employment.
- **Education.** RAIS records eleven education categories. Where we report workforce composition or partition workers into groups, we collapse these into workers without a high school degree, and workers with at least a high school degree. The residualized wage regression instead uses all eleven categories.
- **Tenure.** Months at the establishment as of December, reported in years.
- **Residualized wages.** A worker’s log hourly wage net of the component predicted by that worker’s characteristics. We regress the log wage on a quartic polynomial in age, with the intercept and all four age coefficients free to vary across race $\times$ education $\times$ gender $\times$ year cells. These cells use all eleven education categories recorded in RAIS, the most granular level the data allow. The regression is estimated on the full RAIS sample described in Appendix C.1.5. The establishment-year outcome is the mean of its workers’ residuals, in log points.
- **Clause count.** The number of clauses in the agreement, summed across the clause types defined by *Sistema Mediador*.
- **Wage-equivalent agreement value.** The sum across the 24 *Sistema Mediador* subgroups of an agreement’s clause count in the subgroup times that subgroup’s wage-equivalent value, in log wage points. We take the subgroups’ wage-equivalent values from Table 6 of Lagos (2026). To compute these values, the paper regresses each establishment’s PageRank value (Sorkin, 2018) on its wage premium (following Abowd, Kramarz and Margolis, 1999) and on its clause counts by subgroup. Each subgroup’s coefficient is interpreted as the wage change that leaves workers indifferent to removing one of its clauses, identified from variation within establishments over time.
- **Connectivity.** For each establishment, we count the workers who move between it and any

directly treated establishment over a pair of consecutive years, and divide by its average December employment across the pair. Connectivity is the average of that ratio across the four pre-treatment pairs, 2007–08 through 2010–11. In the regressions the measure is divided by its 90th percentile among untreated establishments in the balanced panel in 2009, so a coefficient reports the effect of moving from zero connectivity to that percentile. The scaling is held fixed across specifications, so estimates are comparable across tables.

- **Group connectivity.** Connectivity computed over a subset of an establishment’s workers. The numerator counts flows between that group’s workers and directly treated establishments. The denominator is the group’s average December employment rather than the establishment’s. The partitions are education, gender, and tenure. Group connectivity is divided by the same firm-level 90th percentile as connectivity, so the two are on a common scale.
- **Counterfactual connectivity.** Expected connectivity under a counterfactual assignment of direct treatment. Cells are defined by industry $\times$ negotiation month, along which the likelihood of direct treatment varies. Holding the number of treated establishments per cell fixed, we reassign the treated label within cells and recompute connectivity from the same pre-reform flows. The measure is the average across 1,000 such reassignments.
- **Per-worker flows.** Defined similarly to connectivity, but counting flows with all establishments in RAIS rather than with directly treated establishments only.
- **Network size.** The number of distinct establishments with which the establishment exchanged at least one worker over the four pre-reform year pairs.
- **Negotiation month.** The calendar month in which agreements for a bargaining unit customarily start and end, known as the *data-base* of the category. It marks when negotiations for the unit should take place. Each agreement reports its base month, and we assign each establishment the mode across its agreements.
- **Industry.** The Brazilian classification of economic activities (CNAE). The time-varying fixed effects use the three-digit classification, which distinguishes, for example, steel from non-steel metalwork.
- **Microregion.** Neighboring municipalities grouped by IBGE into 558 units intended to

capture local labor markets.

Appendix References

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